Review of Derivatives Research (2025) 28:12
https://doi.org/10.1007/s11147-025-09217-4

Stochastic volatility for factor Heath–Jarrow–Morton framework

Artur Sepp[1] · **Parviz Rakhmonov**[2]

Accepted: 26 July 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

We introduce an extension of Factor Heath–Jarrow–Morton (FHJM) framework augmented with a stochastic volatility (SV) driver correlated with factors dynamics. Our model fits the initial yield curve by construction and it is able to produce positive implied volatility skews as observed in market prices of interest rate derivatives. We develop a general framework for analytical valuation of swaptions, interest rate (SOFR) futures, and options on rate futures using moment generating function (MGF) corresponding to the FHJM model with the SV driver. We implement our results to the FHJM model with, first, the CIR SV driver, for which we find the exact solution for model MGF, and, second, for the log-normal SV driver, for which we derive semi-closed-form solution for the MGF based on drift-freezing technique. As a base case, we apply the widespread Nelson–Siegel term structure model for the basis in the FHJM model. We show that Nelson–Siegel model with log-normal SV driver is able to fit accurately market implied volatilities of swaptions across different tenors and expiries and implied volatilities of options on rates futures. Our framework allows for valuation, risk-management and scenario generation for fixed income derivatives including SOFR futures, swaptions, options on rate futures across several tenors and expiries in an arbitrage-free and consistent way.

Keywords Stochastic volatility · Heath–Jarrow–Morton framework · Moment generating function · Nelson–Siegel · SOFR futures · Explosion

The code implementing the analytic solution for swaptions and options of futures based on affine expansion is made available inhttps://github.com/ArturSepp/StochVolModels/tree/main/my_papers/sv_for_factor_hjm.

✉ Parviz Rakhmonov
parviz.msu@gmail.com

Artur Sepp
artursepp@gmail.com

[1] LGT Bank, Zurich, Switzerland

[2] Marex, London, UK

Published online: 02 September 2025

JEL Classification G13 · C63 · C62

1 Introduction

Lyashenko and Goncharov (2023) introduce Factor Heath–Jarrow–Morton (termed as FHJM) framework for modeling the term structure of bond yields by representing it using small number of factors, spanned by selected functional basis [such as Nelson and Siegel (1987)] under statistical $\mathbb{P}$-measure. They augment this representation with auxiliary locally-deterministic factors to ensure the arbitrage-free evolution of the yield curve under the pricing measure $\mathbb{Q}$. The FHJM model has solid advantages, such as intuitiveness of yield curve dynamics, computational efficiency, and absence of arbitrage, while being consistent with the initial term structure by construction.

Yet, an important question for pricing fixed income derivatives is how to model the dependence between curve factors and their volatilities to make the model consistent with implied volatility skews observed in the market. Sepp and Rakhmonov (2023) extend single-factor Cheyette model (which can be viewed as a single-factor FHJM model) to augment the dynamics of the yield curve factor with log-normal SV driver, which is correlated with the factor dynamics. They show that this model can reproduce positive implied volatility skews observed in fixed-income market, which is in contrast to SV models conventionally featuring local volatility and zero correlation with factor dynamics.

In this paper, we extend the FHJM model by incorporating SV dynamics for modeling implied volatilities of swaptions and options on rate futures. As base cases for SV drivers, we consider, first, the affine CIR driver, which allows for exact solutions (up to numerical solutions of ODEs and integrals), and, second, the log-normal SV driver, for which we derive an accurate expansion approach with the same functional form as for the affine driver. The log-normality of the rate volatility for $\mathbb{P}$-dynamics of the one-factor short-term interest rate is well documented by Andersen and Lund (1997) so that for implementations and illustrations we focus on the log-normal SV driver.

1.1 Contributions

We contribute to the literature by introducing stochastic volatility (SV) dynamics for the FHJM model which we term as the FHJM–SV model. We present an analytic approach for valuation of swaptions under the FHJM–SV model using the moment generating function (MGF) associated with this FHJM–SV model. We apply these results for the affine CIR SV model and for log-normal SV model. We show that the CIR SV dynamics may require a strong mean-reversion under the annuity measure. We then focus on the log-normal beta SV model of Karasinski and Sepp (2012) and show that the proposed model is able to fit implied volatilities of swaptions along cross-section of tenors and expiries. Further, we show that the

FHJM–SV model accurately fits implied volatilities of options on SOFR futures. We also derive a closed-form approximation for the futures convexity adjustment within the FHJM–SV framework. Our solution is based on drift-freezing technique and affine expansion approach to obtain a semi-closed-form solution to the MGF, as developed in Sepp and Rakhmonov (2023b) for valuation of vanilla and inverse options.

For practical applications, we incorporate Nelson–Siegel term-structure model (Nelson & Siegel, 1987), which is a particular specification of the FHJM model and which is very popular among market practitioners as well as central banks [see Christensen et al. (2009), Christensen et al. (2011)]. We derive the dynamics of swap rate and rate futures under this model and present extensive tests demonstrating its robustness for pricing applications.

We also propose analytic solution for the SOFR and Eurodollar futures convexity adjustments in the FHJM–SV model which takes into account skew and smile parameters. To our knowledge, this is the first result that incorporates both skew and smile effects within quasi-Gaussian family of models, to which the FHJM model belongs. We derive an exact solution for the convexity adjustment under CIR SV model and an approximate solution under log-normal SV model. In related research, Piterbarg and Renedo (2006) study the futures convexity adjustment in context of LIBOR Market Model (LMM) with stochastic variance, whereas Jäckel and Kawai (2005) consider the displaced lognormal LMM. Mercurio (2017) examines the impact of cap smile in context of displaced log-normal LMM within multi-curve framework.

We additionally study the loss of the martingale property of the futures prices in the FHJM–SV model with log-normal SV driver [see Sin (1998), Andersen and Piterbarg (2007), Bernard et al. (2017), where relevant problem is discussed]. We present sufficient condition for the martingale property of the futures price. Although the problem has been extensively studied, we are not aware of any sufficient conditions of practical applicability which ensure the martingale property for the SOFR futures within SV models. We address this gap within FHJM–SV framework with log-normal SV driver by providing lower bound on quadratic drift which ensures the martingale property for approximate futures rate.

1.2 Organisation

The paper is organised as follows. In Sect. 2, we introduce the FHJM–SV model along with practically important example of Nelson–Siegel basis and of pure exponential (PE) basis. In Sect. 3, we derive the dynamics of the swap rate and of the rate futures under the FHJM–SV model. We further derive analytic solutions for the futures convexity adjustment for the FHJM model with CIR and log-normal SV drivers. We also study the problem of finding convenient sufficient conditions for the futures prices to be martingale in the FHJM–SV model with the log-normal SV driver. In Sect. 4, we present closed-form expression for swaptions and options on rate futures assuming analytic solution for the MGF of the market variable is known. In Sect. 6, we derive affine expansion for the MGF in the FHJM model with log-normal SV driver. In Sect. 7, we calibrate the FHJM–SV model using Nelson–Siegel

basis and log-normal SV driver to market data of swaptions and options on rate futures. We show that the model fits market data very well. Technical proofs are provided in Appendices.

Throughout the text, f denotes a scalar, X and $\vec{\beta}$ denotes a vector, $\mathbf{D}$ denotes a matrix; f_t denote a stochastic variable, $f(t)$ denotes a time-dependent variable, with the same convention for vector and matrix variables. Further, $\top$ denotes the transpose, $\langle \cdot, \cdot \rangle$ denotes scalar product, $\| \cdot \|$ is Euclidian norm, Tr denotes the transpose.

2 Model dynamics

We consider Heath–Jarrow–Morton, see Heath et al. (1992), (HJM) framework which provides generic approach for modeling of the term structure of instantaneous forward rates (as an alternative approach to the HJM framework for instantaneous forward rates, Filipović (2023) proposes arbitrage-free dynamic discount model). Lyashenko and Goncharov (2023) adapt HJM framework for factor-based modeling of the entire forward curve using a small number of Markovian factors. The FHJM model starts with the dynamics of d-dimensional vector X_t of yield curve factors which is specified under the (risk-neutral) $\mathbb{Q}$-measure as follows

$$dX_t = \mathbf{D}X_t dt + \mathbf{\Sigma}_t dW_t^{\mathbb{Q},(0)}, \tag{1}$$

where $\mathbf{D}$ is $d \times d$ generator matrix, $W_t^{\mathbb{Q},(0)}$ is d-dimensional vector of independent Brownian motions under $\mathbb{Q}$ and $\mathbf{\Sigma}_t$ is $d \times d$ volatility matrix.

The instantaneous forward rate $f_t(\tau)$ in FHJM framework admits following factor decomposition

$$f_t(\tau) = B(\tau)X_t + \tilde{B}(\tau)Y_t + \hat{f}_t(\tau), \tag{2}$$

where τ is time-to-maturity, X_t is the main stochastic factor and Y_t is the auxiliary factor. Here $B(\tau)$ is d-dimensional vector of basis functions and $\tilde{B}(\tau)$ is m-dimensional vector of auxiliary basis functions. As shown in Lyashenko and Goncharov (2023), the main basis vector $B(\tau)$ must satisfy

$$B(\tau) = B_0 e^{\mathbf{D}\tau}, \tag{3}$$

where B_0 constant vector and we assume that all eigenvalues κ of $\mathbf{D}$ are such that $\kappa \leq 0$. Initial forward curve $\hat{f}_t(\tau)$ is used to reproduce the observed term structure of forward rates.

The augmented factors dynamics under measure $\mathbb{Q}$ are given by

$$\begin{aligned} dX_t &= \mathbf{D}X_t dt + \mathbf{\Sigma}_t dW_t^{\mathbb{Q},(0)} \\ dY_t &= \tilde{\mathbf{D}}Y_t dt + \Omega_t dt, \end{aligned} \tag{4}$$

where $\tilde{\mathbf{D}}$ is $m \times m$ generator matrix of the auxiliary basis, and Ω_t is a vector process defined as a solution of the equation

$$\tilde{B}(\tau)\Omega_t = B(\tau)\mathbf{\Sigma}_t\mathbf{\Sigma}_t^\top\left(\int_0^\tau B(u)\,du\right)^\top. \tag{5}$$

Lyashenko and Goncharov (2023) show that the generating matrix $\tilde{\mathbf{D}}$ of the auxiliary basis $\tilde{B}(\tau)$ can be chosen having spectrum $\tilde{\Lambda}$, where

$$\tilde{\Lambda} = \Lambda \cup \{\Lambda + \Lambda\}, \quad \Lambda + \Lambda := \{\lambda_i + \lambda_j \mid \lambda_i, \lambda_j \in \Lambda\}, \tag{6}$$

where Λ is the spectrum of the matrix D. Thus, FHJM framework allows to judiciously specify generator matrix $\tilde{\mathbf{D}}$ and extended basis $\tilde{B}(\tau)$ for the auxiliary process Y_t when using the exponential basis in Eq. (3).

We emphasize that the FHJM framework relies on Musiela parametrization of the Heath–Jarrow–Morton dynamics introduced in Musiela (1993). Thus, the FHJM model is formulated in terms of time to maturity or tenor variable τ, $\tau = T - t$, where t is current time and T is time of maturity, rather than in terms of T. Both parametrizations are in fact equivalent and can be related using, e.g. Itô–Wentzell formula.[1] However, working with tenor τ dimension simplifies our derivations and enforces time-homogeneity in the model.

2.1 Stochastic volatility

The key purpose of adding the SV dynamics is to model interest rate volatility skews observed in the market data. We account that using d-factor volatility for each of factors X_t may not be well-identifiable because factors are not observable, unlike the forward rates. In Appendix 1 we discuss the specification of the volatility matrix $\mathbf{\Sigma}_t$ in dynamics (4) for arbitrary basis $\mathbf{B}(\tau)$. We show that $\mathbf{\Sigma}_t$ must be independent from X_t to avoid explosions so that, conditional on $\mathbf{\Sigma}_t$, the factor dynamics are Gaussian. We present a statistical method to estimate $\mathbf{\Sigma}_t$ using the correlation between benchmark yields and their instantaneous volatilities.

Thus, we consider the following scalar-based structure for the volatility matrix

$$\mathbf{\Sigma}_t = \sigma_t\mathbf{C}(t), \tag{7}$$

where $\mathbf{C}(t)$ is $d \times d$ time-dependent matrix which specifies the instantaneous volatility of factors, and σ_t is one-dimensional SV process. We follow Andersen and Piterbarg (2010) by considering the single volatility σ_t for scaling of the volatility matrix in Eq. (7) which is sufficient for pricing single-rate vanilla derivatives, the same approach is used in Duchitskii and Piterbarg (2024). We shall leave the extension to multi-dimensional SV setting for future research.

Without loss of generality we assume that $\sigma_0 = 1$ so that σ_t incorporates stochastic fluctuations around unity. Let us consider a general model using the following dynamics of the SV driver under $\mathbb{Q}$ measure:

$$d\sigma_t = \nu(\sigma_t)dt + \epsilon(\sigma_t)\left(\vec{\beta}(t)^\top\, dW_t^{\mathbb{Q},(0)} + \beta_0(t)dw_t^{\mathbb{Q},(1)}\right), \quad \sigma_0 = \sigma, \tag{8}$$

[1] We wish to thank Vladimir Lucic for a helpful suggestion.

where $\nu(\sigma_t)$ is the volatility drift and $\epsilon(\sigma_t)$ is the volatility of volatility; $\vec{\beta}(t)$ is d-dimensional beta coefficient which defines factors loadings for the volatility dynamics projected by stochastic changes in factor dynamics given by $W_t^{\mathbb{Q},(0)}$, $w_t^{\mathbb{Q},(1)}$ is independent one-dimensional Brownian motion with volatility-of-volatility $\beta_0(t)$. Factor loadings $\vec{\beta}(t)$ can be identified statistically through multi-variate projection of dX_t into $d\sigma_t$. In summary, we propose the following dynamics of the FHJM model with SV factor.

Assumption 2.1 (Dynamics of the FHJM–SV model) The joint dynamics of factors in the FHJM–SV model under the risk-neutral measure $\mathbb{Q}$ are given by

$$\begin{aligned} dX_t &= \mathbf{D}X_t dt + \sigma_t \mathbf{C}(t) dW_t^{\mathbb{Q},(0)}, \quad X_0 = X_0 \\ dY_t &= \tilde{\mathbf{D}}Y_t dt + \sigma_t^2 \tilde{\Omega}_t dt, \quad Y_0 = 0 \\ d\sigma_t &= \nu(\sigma_t)dt + \epsilon(\sigma_t)\Big(\vec{\beta}(t)^\top dW_t^{\mathbb{Q},(0)} + \beta_0(t) dw_t^{\mathbb{Q},(1)}\Big), \quad \sigma_0 = \sigma, \end{aligned} \tag{9}$$

where vector $\tilde{\Omega}_t$ is defined by

$$\tilde{B}(\tau)\tilde{\Omega}_t = B(\tau)\mathbf{C}(t)\mathbf{C}(t)^\top \left(\int_0^\tau B(u)\, du\right)^\top. \tag{10}$$

As to practical specifications of SV drivers for model dynamics (9), we consider the following SV models. First, we introduce the CIR dynamics for $\sigma_t = \sqrt{V_t}$, with stochastic variance V_t driven by

$$dV_t = \kappa\big(\theta - V_t\big)dt + \sqrt{V_t}\Big(\vec{\beta}(t)^\top dW_t^{\mathbb{Q},(0)} + \beta_0(t)\, dw_t^{\mathbb{Q},(1)}\Big), \quad V_0 = V, \tag{11}$$

where $\kappa > 0$ is the mean-reversion speed and $\theta > 0$ is the long-run variance.

Second, we consider the log-normal SV beta model with quadratic drift introduced by Sepp and Rakhmonov (2023b)

$$d\sigma_t = \big(\kappa_1 + \kappa_2\sigma_t\big)(\theta - \sigma_t)dt + \sigma_t\Big(\vec{\beta}(t)^\top dW_t^{\mathbb{Q},(0)} + \beta_0(t) dw_t^{\mathbb{Q},(1)}\Big), \quad \sigma_0 = \sigma, \tag{12}$$

where κ_1 and κ_2 are linear and quadratic mean-reversion speeds, respectively; θ is the long-run level of volatility. We make the following assumption about the parameters of process σ_t.

Assumption 2.2 Model parameters κ_1, κ_2, θ in Eq. (12) satisfy the following conditions

$$\kappa_1 \geq 0, \quad \kappa_2 \geq 0, \quad \theta > 0. \tag{13}$$

Theorem 2.1 *We assume an FHJM model with basis $B(\tau)$ and with the dynamics of model factors $\{X_t, Y_t\}$ under $\mathbb{Q}$ given in Eq. (9). Instantaneous forward rates $f_t(\tau)$ are then given by*

$$f_t(\tau) = B(\tau)X_t + \tilde{B}(\tau)Y_t + \hat{f}_t(\tau), \quad \hat{f}_t(\tau) = f_0(t+\tau). \tag{14}$$

Prices of zero-coupon bonds are given by

$$P(t, t+\tau) \equiv P(t, t+\tau; X_t, Y_t) = \frac{P(0, t+\tau)}{P(0,t)} \exp\left(-B_P(\tau)X_t - \tilde{B}_P(\tau)Y_t\right), \tag{15}$$

where

$$B_P(\tau) := \int_0^\tau B(u)\,du, \quad \tilde{B}_P(\tau) := \int_0^\tau \tilde{B}(u)\,du. \tag{16}$$

Proof We only need to prove Eq. (15). Indeed, by integrating Eq. (14), we obtain

$$\begin{aligned} P(t, t+\tau) &= \exp\left(-\int_0^\tau f_t(u)\,du\right) = \exp\left(-\int_0^\tau \left(f_0(t+u) + B(u)X_t + \tilde{B}(u)Y_t\right)du\right) \\ &= \frac{P(0,t+\tau)}{P(0,t)} \exp\left(-\int_0^\tau B(u)\,du\,X_t - \int_0^\tau \tilde{B}(u)\,du\,Y_t\right). \end{aligned}$$

□

We note that model associated with FHJM–SV dynamics (9) and bond formula (15) exhibits the so-called unspanned stochastic volatility, see Collin-Dufresne and Goldstein (2002).

2.2 Single-factor exponential basis

As an introduction, we consider a single factor FHJM model with the one-dimensional basis

$$B(\tau) = \left(e^{-\lambda\tau}\right), \tag{17}$$

where λ, $\lambda > 0$, is the mean-reversion rate. By applying Theorem 2.1 we obtain the following result.

Corollary 2.1 *Consider FHJM model with basis given by* (17)*, where the dynamics of model factors X_t, Y_t are given in Eq.* (9)*. Then, generating matrix of the main basis $B(\tau)$, auxiliary basis $\tilde{B}(\tau)$ and vector $\tilde{\Omega}_t$ are given by*[2]

$$\mathbf{D} = (-\lambda), \quad \tilde{\mathbf{D}} = \begin{pmatrix} -\lambda & 0 \\ 0 & -2\lambda \end{pmatrix}, \quad \tilde{B}(\tau) = \left(e^{-\lambda\tau}, e^{-2\lambda\tau}\right), \quad \tilde{\Omega}_t = \frac{1}{\lambda}\left(c(t)^2, -c(t)^2\right). \tag{18}$$

where $c(t)$ is the deterministic volatility (or a leverage factor relative to the stochastic volatility driver σ_t).

[2] To preserve consistency with our notations, since volatility matrix $\mathbf{C}(t)$ is 1×1 matrix, we denote it by $c(t)$.

The term structure of instantaneous forward rates $f_t(\tau)$ are given by

$$f_t(\tau) = B(\tau)X_t + \tilde{B}(\tau)Y_t + \hat{f}_t(\tau), \quad \hat{f}_t(\tau) = f_0(t+\tau).$$

Prices of zero-coupon bonds are given by

$$P(t,t+\tau) = P(t,t+\tau;X_t,Y_t) = \frac{P(0,t+\tau)}{P(0,t)} \exp\left(-B_P(\tau)X_t - \tilde{B}_P(\tau)Y_t\right), \quad (19)$$

where

$$B_P(\tau) = \frac{1-e^{-\lambda\tau}}{\lambda}, \quad \tilde{B}_P(\tau) = \left(\frac{1-e^{-\lambda\tau}}{\lambda}, \frac{1-e^{-2\lambda\tau}}{2\lambda}\right)^\top.$$

Straightforward calculations show that when $\lambda = 0$, main basis $B(\tau)$, auxiliary basis $\tilde{B}(\tau)$ and vector $\tilde{\Omega}_t$ degenerate into

$$\mathbf{D} = (0), \quad \tilde{\mathbf{D}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B(\tau) = 1, \quad \tilde{B}(\tau) = (1,\tau), \quad \tilde{\Omega}_t = \left(0, c(t)^2\right)^\top. \quad (20)$$

Remark. As seen, when factor dynamics are Gaussian, i.e. $\sigma_t \equiv 1$, this model corresponds to continuous time Hull-White and Ho-Lee short rate models when $\lambda > 0$ and $\lambda = 0$, respectively.

Let us we draw the distinction between the FHJM and Cheyette models. Although both the FHJM and Cheyette model belong to the quasi-Gaussian family of term-structure models, parametrization of state variables are different in these models. Under the FHJM model, the dynamics of main driver X and locally deterministic variable Y are formally independent, while in conventional Cheyette model the auxiliary variable Y is present in the drift of X. However, in both models the solution to the instantaneous forward rate must be unique, see Lyashenko and Goncharov (2023) for details.

2.3 Nelson–Siegel basis

While we develop generic solution for the FHJM–SV dynamics (9), we apply Nelson–Siegel basis, see Nelson and Siegel (1987), as a viable application, which is widely used in practice [see Filipović (1999), Diebold and Li (2006), Christensen et al. (2009, 2011), among others].

Nelson–Siegel basis is specified by

$$B(\tau) = \left(1, e^{-\lambda\tau}, \tau e^{-\lambda\tau}\right), \quad (21)$$

where λ is interpreted as a mean reversion rate. As highlighted in Diebold and Li (2006), λ controls the tenor at which the curvature factor achieves its maximum.

For Nelson–Siegel basis, the generating matrix representing Exponential-Polynomial (EP) basis [see Filipović (2000)] in Eq. (3) is specified by

$$\mathbf{D} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\lambda & 1 \\ 0 & 0 & -\lambda \end{pmatrix}. \tag{22}$$

Furthermore, the auxiliary basis can also be represented through the EP family

$$\tilde{B}(\tau) = \left(1, \tau; e^{-\lambda\tau}, \tau e^{-\lambda\tau}, \frac{\tau^2}{2} e^{-\lambda\tau}; e^{-2\lambda\tau}, \tau e^{-2\lambda\tau}, \frac{\tau^2}{2} e^{-2\lambda\tau}\right), \tag{23}$$

where

$$\tilde{\mathbf{D}} = \begin{pmatrix} 0 & 1 & & & & & & & \\ & 0 & & & & & & & \\ & & -\lambda & 1 & & & & & \\ & & & -\lambda & 1 & & & & \\ & & & & -\lambda & & & & \\ & & & & & -2\lambda & 1 & & \\ & & & & & & -2\lambda & 1 & \\ & & & & & & & -2\lambda & \\ & & & & & & & & -2\lambda \end{pmatrix}. \tag{24}$$

We highlight that matrix $\tilde{\mathbf{D}}$ is block diagonal.

Let us find the vector $\tilde{\Omega}_t$ from Eq. (10): if we denote

$$\tilde{\Omega}_t := \left(\tilde{\Omega}_t^1, \tilde{\Omega}_t^2, \ldots, \tilde{\Omega}_t^8\right)^\top, \quad \mathbf{M}(t) := \mathbf{C}(t)\mathbf{C}(t)^\top, \quad \mathbf{M}(t) = \{m_{kj}(t)\}_{k,j=1,2,3},$$

then by rearranging the terms we find that

$$\begin{aligned} \tilde{\Omega}_t^1 &= \frac{m_{12}}{\lambda} + \frac{m_{13}}{\lambda^2}, & \tilde{\Omega}_t^2 &= m_{11}, \\ \tilde{\Omega}_t^3 &= -\frac{m_{12}}{\lambda} - \frac{m_{13}}{\lambda^2} + \frac{m_{22}}{\lambda} + \frac{m_{23}}{\lambda^2}, & \tilde{\Omega}_t^4 &= m_{12} - \frac{m_{13}}{\lambda} + \frac{m_{23}}{\lambda} + \frac{m_{33}}{\lambda^2}, \\ \tilde{\Omega}_t^5 &= 2m_{13}, & \tilde{\Omega}_t^6 &= -\frac{m_{22}}{\lambda} - \frac{m_{23}}{\lambda^2}, \\ \tilde{\Omega}_t^7 &= -\frac{2m_{23}}{\lambda} - \frac{m_{33}}{\lambda^2}, & \tilde{\Omega}_t^8 &= -\frac{2m_{33}}{\lambda}, \end{aligned} \tag{25}$$

where we drop the argument t in $m_{kj}(t)$ for notational convenience.

Corollary 2.2 *Consider Nelson–Siegel term-structure model, which is an FHJM model with basis $B(\tau)$ given by (21). Then the dynamics of model factors $\{X_t, Y_t\}$ are given by (9), where generating matrix of the main basis $B(\tau)$, auxiliary basis $\tilde{B}(\tau)$ and vector $\tilde{\Omega}_t$ are given by Eqs. (22), (24) and (25), respectively.*

The term structure of the instantaneous forward rates $f_t(\tau)$ are then given by

$$f_t(\tau) = B(\tau)X_t + \tilde{B}(\tau)Y_t + \hat{f}_t(\tau), \quad \hat{f}_t(\tau) = f_0(t+\tau).$$

Prices of zero-coupon bonds are given by

$$P(t,t+\tau) = P(t,t+\tau;X_t,Y_t) = \frac{P(0,t+\tau)}{P(0,t)} \exp\left(-B_P(\tau)X_t - \tilde{B}_P(\tau)Y_t\right), \quad (26)$$

where

$$B_P(\tau) = \left(\tau, \frac{1-e^{-\lambda\tau}}{\lambda}, \frac{1-e^{-\lambda\tau}(1+\lambda\tau)}{\lambda^2}\right),$$

$$\tilde{B}_P(\tau) = \left(\tau, \frac{\tau^2}{2}; \frac{1-e^{-\lambda\tau}}{\lambda}, \frac{1-e^{-\lambda\tau}(1+\lambda\tau)}{\lambda^2}, \frac{1-e^{-\lambda\tau}(1+\lambda\tau+\frac{1}{2}\lambda^2\tau^2)}{\lambda^3}; \right.$$
$$\left. \frac{1-e^{-2\lambda\tau}}{2\lambda}, \frac{1-e^{-2\lambda\tau}(1+2\lambda\tau)}{4\lambda^2}, \frac{1-e^{-2\lambda\tau}(1+2\lambda\tau+2\lambda^2\tau^2)}{8\lambda^3}\right).$$

Proof Follows from Theorem 2.1. □

We further consider multi-dimensional exponential basis in Appendix 2 as an alternative to Nelson–Siegel basis.

3 Dynamics of market instruments

In this section, we derive the dynamics of swap rates and the dynamics of rate futures under the FHJM–SV model (9). We extend the setup developed for the single-factor model in Sepp and Rakhmonov (2023a) to the FHJM–SV framework.

We introduce the tenor structure $0 \le T_0 < T_1 < \cdots < T_N$ and define annuity factor A by

$$A(t) \equiv A(t;X_t,Y_t) = \sum_{n=0}^{N-1} \tau_n P(t,T_{n+1};X_t,Y_t), \quad \tau_n = T_{n+1} - T_n, \quad (27)$$

with $A(0)$ determined by the term structure of interest rates observed currently. We specify the forward swap rate $S(t)$ by

$$S(t) \equiv S(t;X_t,Y_t) = \frac{P(t,T_0;X_t,Y_t) - P(t,T_N;X_t,Y_t)}{A(t;X_t,Y_t)}, \quad t \le T_0. \quad (28)$$

For future developments, we introduce:

$$L(t) = \ln A(t), \quad L_X(t) \equiv \nabla_X L(t) = \frac{1}{A(t)} \nabla_X A(t), \quad (29)$$

where $\nabla_X = \left(\partial_{x_1}, \ldots, \partial_{x_d}\right)$ denotes gradient operator.

We compute closed-form expressions for $L_X(t,X,Y)$ and $\nabla_X S(t,X,Y)$ using bond-reconstruction formula (15) as follows, using shorthand $P(t,T) = P(t,T;X_t,Y_t)$:

$$
\begin{aligned}
L_X(t;X_t,Y_t) &= -\frac{1}{A(t)}\sum_{n=0}^{N-1}\tau_n P(t,T_{n+1})B_P(T_{n+1}-t)^\top,\\
\nabla_X S(t;X_t,Y_t) &= \frac{1}{A(t)}\left(-P(t,T_0)B_P(T_0-t)^\top + P(t,T_N)B_P(T_N-t)^\top\right)+\\
&\quad + \frac{P(t,T_0)-P(t,T_N)}{A(t)^2}\left[\sum_{n=0}^{N-1}\tau_n P(t,T_{n+1})B_P(T_{n+1}-t)^\top\right].
\end{aligned}
\tag{30}
$$

3.1 State dynamics under annuity measure

We consider annuity measure $\mathbb{Q}^A$ by choosing annuity $A(t)$ as numeraire. To keep the dynamics of the volatility process generic, we assume that condition $\mathbb{Q}\sim\mathbb{Q}^A$ is satisfied. And discuss sufficient conditions in case of log-normal SV and CIR drivers later on.

Theorem 3.1 *Assume that $\mathbb{Q}\sim\mathbb{Q}^A$. Then, the dynamics of state variables $\{X_t, Y_t, \sigma_t\}$ under the annuity measure $\mathbb{Q}^A$ for model dynamics (9) are given by*

$$
\begin{aligned}
dX_t &= \mathbf{D}X_t dt + \sigma_t^2\mathbf{C}(t)\mathbf{C}(t)^\top L_X(t)dt + \sigma_t\mathbf{C}(t)\,dW_t^{A,(0)} \quad X_0 = X_0,\\
dY_t &= \tilde{\mathbf{D}}Y_t dt + \sigma_t^2\tilde{\Omega}_t dt, \quad Y_0 = 0\\
d\sigma_t &= \left[\nu(\sigma_t) + \sigma_t\epsilon(\sigma_t)\boldsymbol{\beta}(t)\mathbf{C}(t)^\top L_X(t)\right]dt + \epsilon(\sigma_t)\left(\vec{\beta}(t)^\top dW_t^{A,(0)} + \beta_0(t)dw_t^{A,(1)}\right),
\end{aligned}
\tag{31}
$$

where $L_X(t)$ is defined in Eq. (29).

Proof See Appendix 3.1. □

Corollary 3.1 *Using dynamics in Eq. (12), we obtain*

$$
\begin{aligned}
dX_t &= \mathbf{D}X_t dt + \sigma_t^2\mathbf{C}(t)\mathbf{C}(t)^\top L_X(t)dt + \sigma_t\mathbf{C}(t)dW_t^{A,(0)}, \quad X_0 = X_0\\
dY_t &= \tilde{\mathbf{D}}Y_t dt + \sigma_t^2\tilde{\Omega}_t dt, \quad Y_0 = 0\\
d\sigma_t &= \left[\kappa_1\theta - (\kappa_1-\kappa_2\theta)\sigma_t - \left(\kappa_2 - \vec{\beta}(t)^\top\mathbf{C}(t)^\top L_X(t)\right)\sigma_t^2\right]dt+\\
&\quad + \sigma_t\vec{\beta}(t)^\top dW_t^{A,(0)} + \beta_0(t)\sigma_t\,dw_t^{A,(1)}.
\end{aligned}
\tag{32}
$$

As shown in the proof, see Appendix 3.1, due to the correction in the quadratic speed of mean-reversion, we must impose the condition

$$
\kappa_2 > \vec{\beta}(t)^\top\mathbf{C}(t)^\top L_X(t),
\tag{33}
$$

so that the volatility process σ_t in (32) admits strong solution and to ensure equivalence of measures $\mathbb{Q}\sim\mathbb{Q}^A$, which is required for the application of Girsanov theorem in Corollary 3.1.

As element-wise $B_P(\tau) > 0$ for $\tau > 0$, each element of the vector $L_X(t)$ is positive. Nevertheless, elements of $\mathbf{C}(t)^\top L_X(t)$ can be of any sign, hence $\vec{\beta}(t)^\top \mathbf{C}(t)^\top L_X(t)$ can be of any sign as well.

Corollary 3.2 *Using dynamics CIR SV dynamics in Eq. (11), we obtain*

$$\begin{aligned} dX_t &= \mathbf{D} X_t dt + V_t \mathbf{C}(t)\mathbf{C}(t)^\top L_X(t) dt + \sqrt{V_t}\mathbf{C}(t) dW_t^{A,(0)}, \quad X_0 = X_0 \\ dY_t &= \tilde{\mathbf{D}} Y_t dt + V_t \tilde{\Omega}_t dt, \quad Y_0 = 0 \\ dV_t &= \left[\kappa\theta - \left(\kappa - \boldsymbol{\beta}(t)\mathbf{C}(t)^\top L_X(t)\right) V_t\right] dt + \sqrt{V_t}\left(\vec{\beta}(t)^\top dW_t^{A,(0)} + \beta_0(t)\, dw_t^{A,(1)}\right). \end{aligned} \tag{34}$$

As seen, the change of measure from $\mathbb{Q}$ to $\mathbb{Q}^A$ results in correction of the mean reversion speed. To avoid the situation of having mean repelling variance process in model (34), one must impose a restriction $\kappa > \vec{\beta}(t)^\top \mathbf{C}(t)^\top L_X(t)$.

3.2 Dynamics of swap rate

We consider the model swap rate $S(t) \equiv S(t; X_t, Y_t)$ defined in (28). Since $S(t)$ has zero drift under $\mathbb{Q}^A$, we have

$$\begin{aligned} dS(t) &= \frac{\partial S(t)}{\partial t} dt + \left(\nabla_X S(t)\right)^\top dX_t + \left(\nabla_Y S(t)\right)^\top dY_t + \frac{1}{2} Tr\left\{\boldsymbol{\Sigma}_t^\top \nabla_X^2 S(t) \boldsymbol{\Sigma}_t\right\} dt = \\ &= \left(\nabla_X S(t)\right)^\top \boldsymbol{\Sigma}_t\, dW_t^{A,(0)}. \end{aligned} \tag{35}$$

where $\nabla_X S(t)$ is defined in Eq. (30) and $\boldsymbol{\Sigma}_t$ is defined in Eq. (7). Accordingly, we obtain the following result.

Proposition 3.1 *The dynamics of forward swap rate $S(t)$ under annuity measure $\mathbb{Q}^A$ is given by*

$$dS(t, X_t, Y_t) = \sigma_t \left(\nabla_X S(t)\right)^\top \mathbf{C}(t)\, dW_t^{A,(0)}, \tag{36}$$

where dynamics of state variables $\{X_t, Y_t, \sigma_t\}$ are given in Eq. (31).

Gradients $\nabla_X \ln A(t)$ and $\nabla_X S(t)$ are analytical functions of state drivers $\{X_t, Y_t\}$, yet they are non-linear, which hinders analytical tractability for model calibration. For analytical tractability, we replace path trajectories $\{X_t, Y_t\}$ in function $A(t)$ by the path of their deterministic expected values under the annuity measure

$$X_t \to \bar{X}_t = \mathbb{E}^A(X_t), \quad Y_t \to \bar{Y}_t = \mathbb{E}^A(Y_t), \tag{37}$$

which are computed by applying expected value operator to both sides of Eq. (31). This technique is also known as 'freezing the drift' (Brigo & Mercurio, 2007) which is applied for deriving analytically tractable LIBOR market models.

As a result, we obtain following system of ODEs for the expected values $\{\bar{X}_t, \bar{Y}_t\}$ under the dynamics (31):

$$\begin{aligned} d\bar{X}_t &= \mathbf{D}\bar{X}_t dt + \mathbf{C}(t)\mathbf{C}(t)^\top \mathbb{E}^A\left[\sigma_t^2 L_X(t)\right] dt, \\ d\bar{Y}_t &= \tilde{\mathbf{D}}\bar{Y}_t dt + \tilde{\Omega}_t \mathbb{E}^A[\bar{\sigma}_t^2] dt. \end{aligned} \tag{38}$$

Finally, we propose the following dynamics of the approximate swap rate.

Proposition 3.2 (State-independent swap rate s_t under FHJM–SV dynamics (9)) *Dynamics of state-independent swap rate s_t are given by*

$$\begin{aligned} ds_t &= \sigma_t \times a(t)^\top \, dW_t^{A,(0)} \\ d\sigma_t &= \left[\nu(\sigma_t) + \sigma_t \epsilon(\sigma_t)\beta_2(t)\right] dt + \epsilon(\sigma_t)\vec{\beta}(t)^\top \, dW_t^{A,(0)} + \beta_0(t)\epsilon(\sigma_t) dw_t^{A,(1)}, \end{aligned} \tag{39}$$

where we introduce the time-dependent level of the volatility $a(t)$ and volatility beta $\beta_2(t)$ below

$$a(t) = \mathbf{C}(t)^\top \nabla_X S(t, \bar{X}_t, \bar{Y}_t), \;\; \beta_2(t) = \vec{\beta}(t)^\top \mathbf{C}(t)^\top L_X(t, \bar{X}_t, \bar{Y}_t). \tag{40}$$

with $\bar{X}_t, \bar{Y}_t$ being solutions to Eq. (38).

Corollary 3.3 *The dynamics of s_t with CIR SV dynamics (11) are given by*

$$\begin{aligned} ds_t &= \sqrt{V_t} \times a(t)^\top \, dW_t^{A,(0)} \\ dV_t &= \left[\kappa\theta - \left(\kappa - \beta_2(t)\right) V_t\right] dt + \sqrt{V_t}\left(\vec{\beta}(t)^\top \, dW_t^{A,(0)} + \beta_0(t) \, dw_t^{A,(1)}\right). \end{aligned} \tag{41}$$

Hereby $\bar{X}_t$ and $\bar{Y}_t$ are computed as follows

$$\begin{aligned} d\bar{X}_t &= \mathbf{D}\bar{X}_t dt + \bar{V}_t \mathbf{C}(t)\mathbf{C}(t)^\top L_X(t, \bar{X}_t, \bar{Y}_t) \, dt, \\ d\bar{Y}_t &= \tilde{\mathbf{D}}\bar{Y}_t dt + \bar{V}_t \tilde{\Omega}_t \, dt, \\ d\bar{V}_t &= \left[\kappa\theta - \left(\kappa - \boldsymbol{\beta}(t)\mathbf{C}(t)^\top L_X(t, \bar{X}_t, \bar{Y}_t)\right)\bar{V}_t\right] dt. \end{aligned} \tag{42}$$

Corollary 3.4 *The dynamics of s_t with log-normal SV dynamics (12) are given by*

$$\begin{aligned} ds_t &= \sigma_t \times a(t)^\top \, dW_t^{A,(0)}, \\ d\sigma_t &= \left[\kappa_1\theta - (\kappa_1 - \kappa_2\theta)\sigma_t - (\kappa_2 - \beta_2(t))\sigma_t^2\right] dt + \sigma_t \vec{\beta}(t)^\top \, dW_t^{A,(0)} + \sigma_t \beta_0(t) \, dw_t^{A,(1)}. \end{aligned} \tag{43}$$

Hereby, expected values $\{\bar{X}_t, \bar{Y}_t\}$ are computed using log-normal SV dynamics (9). Taking expected values in (31), we obtain that expected values $\{\bar{X}_t, \bar{Y}_t\}$ satisfy the system of ODEs

$$
\begin{aligned}
d\bar{X}_t &= \mathbf{D}\bar{X}_t dt + \mathbf{C}(t)\mathbf{C}(t)^\top \times \mathbb{E}^A\left[\sigma_t^2 L_X(t)\right] dt, \\
d\bar{Y}_t &= \tilde{\mathbf{D}}\bar{Y}_t dt + \tilde{\Omega}_t \times \mathbb{E}^A\left[\sigma_t^2\right] dt, \\
d\bar{\sigma}_t &= \left(\kappa_1\theta - (\kappa_1 - \kappa_2\theta)\bar{\sigma}_t - \kappa_2\mathbb{E}^A[\sigma_t^2]\right) dt + \vec{\beta}(t)^\top \mathbf{C}(t)^\top \mathbb{E}^A\left[\sigma_t^2 L_X(t)\right] dt.
\end{aligned} \tag{44}
$$

Following similar approximation proposed in Sepp and Rakhmonov (2023a) for the single-factor model, we expand the expressions inside the expectations by ignoring higher order terms as follows:

$$
\begin{aligned}
\mathbb{E}^A\left[\sigma_t^2 L_X(t)\right] &\approx \bar{\sigma}_t^2 L_X(t, \bar{X}_t, \bar{Y}_t), \\
\mathbb{E}^A\left[\sigma_t^2\right] &\approx \bar{\sigma}_t^2.
\end{aligned} \tag{45}
$$

While $\mathbb{E}^A\left[\sigma_t^2\right]$ can be computed using a higher order discretization scheme, we have found no advantages in using such a scheme. Coupled system in Eqs. (44)–(45) or in Eq. (42) is solved numerically using standard ODE solvers.

3.3 Forward and futures rates

We first introduce some notation for overnight-based futures, following Lyashenko and Mercurio (2019), among others. We let T_S, T_E denote start and end date, $T_S < T_E$ and $\{t_n\}_{n=1}^N$ be the collection of business days over $[T_S, T_E]$. Going forward, we consider only SOFR futures, although our results are valid for futures linked to any overnight rate (ON). By definition, SOFR futures settlement at time T_E is based on daily SOFR, compounded over $[T_S, T_E]$ as follows

$$
\hat{F}^{\text{ON}}(T_S, T_E) = \frac{1}{\Delta}\left[\prod_{n=1}^{N}\left(1 + \Delta_n R_n\right) - 1\right] \tag{46}
$$

where Δ denotes year fraction for $[T_S, T_E]$, R_n is SOFR rate over $[t_{n-1}, t_n]$ and Δ_n is the associated day-count fraction. Following standard market practice, we approximate discrete compounding by continuous one and consider

$$
F^{\text{ON}}(T_S, T_E) := \frac{1}{\Delta}\left(e^{\int_{T_S}^{T_E} r(t)\,dt} - 1\right), \tag{47}
$$

where $r(t)$ is the instantaneous risk-free rate, linked to the SOFR rate. This approximation offers significant computational and notational advantage. In practice, the difference between continuous and discrete daily compounding is extremely small, thus neglected.

We first consider futures rate $F_t^{\text{ON}} := F^{\text{ON}}(t; T_S, T_E)$ with settlement value $F^{\text{ON}}(T_S, T_E)$ given in Eq. (47), $t \le T_E$, $\Delta = T_E - T_S$. It is well-known, see Hunt and Kennedy (2004), that futures value $F^{\text{ON}}(t; T_S, T_E)$ is computed as follows

$$
F^{\text{ON}}(t; T_S, T_E) := \mathbb{E}^{\mathbb{Q}}\left[F^{\text{ON}}(T_S, T_E)\middle|\mathcal{F}_t\right] = \frac{1}{\Delta}\left(\mathbb{E}^{\mathbb{Q}}\left[e^{\int_{T_S}^{T_E} r(t)\,dt}\middle|\mathcal{F}_t\right] - 1\right), \tag{48}
$$

where expectation is taken under risk-neutral measure $\mathbb{Q}$.

In addition to settlement value in Eq. (47), based on compounded SOFR rate, let us consider Eurodollar futures that settles at time T_S into value

$$F^{\mathrm{ED}}(T_S, T_E) = \frac{1}{\Delta}\big(P(T_S, T_E)^{-1} - 1\big). \tag{49}$$

Thus, futures rate at time $t \le T_S$ with settlement value $F^{\mathrm{ED}}(T_S, T_E)$ in Eq. (49), equals to

$$F^{\mathrm{ED}}(t;T_S, T_E) := \mathbb{E}^{\mathbb{Q}}\Big[F^{\mathrm{ED}}(T_S, T_E)\Big|\mathcal{F}_t\Big] = \frac{1}{\Delta}\Big(\mathbb{E}^{\mathbb{Q}}\Big[P(T_S, T_E)^{-1}\Big|\mathcal{F}_t\Big] - 1\Big). \tag{50}$$

As the value of futures rate is model-dependent, we present the explicit formula for $F^{\mathrm{ON}}(t;T_S, T_E)$ and $F^{\mathrm{ED}}(t;T_S, T_E)$ in Theorem 3.2 to derive the convexity adjustment formula for FHJM–SV model (9). We note that since SOFR futures are backward looking, it is more convenient to work with concept of extended zero-coupon bond as in Lyashenko and Mercurio (2019).

Theorem 3.2 *We consider general dynamics of FHJM–SV model* (9) *and let* $0 \le t \le T_E$. *Time t value of SOFR futures* $F^{\mathrm{ON}}(t;T_S, T_E)$ *is given by*

$$\begin{aligned} F^{\mathrm{ON}}(t;T_S, T_E) &= \frac{1}{\Delta}\left(\frac{P(t, T_S)}{P(t, T_E)}c^{\mathrm{ON}}(t) - 1\right), \\ c^{\mathrm{ON}}(t) &= \mathbb{E}^{\mathbb{Q}}\left\{ e^{B(0)\int_{T_S\wedge t}^{T_E} X_u\,du + \tilde{B}(0)\int_{T_S\wedge t}^{T_E} Y_u\,du}\middle|\mathcal{F}_t\right\} \\ &\quad \times e^{-\left(B_P(T_E - T_E\vee t) - B_P(T_S - T_S\vee t)\right)X_t - \left(\tilde{B}_P(T_E - T_E\vee t) - \tilde{B}_P(T_S - T_S\vee t)\right)Y_t}. \end{aligned} \tag{51}$$

Let $0 \le t \le T_S$. *Time t value of Eurodollar futures* $F^{\mathrm{ED}}(t;T_S, T_E)$ *is given by*

$$\begin{aligned} F^{\mathrm{ED}}(t;T_S, T_E) &= \frac{1}{\Delta}\left(\frac{P(t, T_S)}{P(t, T_E)}c^{\mathrm{ED}}(t) - 1\right), \\ c^{\mathrm{ED}}(t) &= \mathbb{E}^{\mathbb{Q}}\left\{ e^{B_P(\Delta)X_{T_S} + \tilde{B}_P(\Delta)Y_{T_S}}\middle|\mathcal{F}_t\right\} e^{-\left(B_P(T_E - t) - B_P(T_S - t)\right)X_t - \left(\tilde{B}_P(T_E - t) - \tilde{B}_P(T_S - t)\right)Y_t}. \end{aligned} \tag{52}$$

Proof Let us prove (52) first. We use the following relationship, derived by integrating (14):

$$\int_0^{\Delta} f_{T_S}(\tau)\,d\tau = \int_0^{\Delta} \hat{f}_{T_S}(\tau)\,d\tau + B_P(\Delta)X_{T_S} + \tilde{B}_P(\Delta)Y_{T_S}, \tag{53}$$

where $B_P(\Delta)$, $\tilde{B}_P(\Delta)$ are defined in (16). For $t \le T_S$ we have, on the other hand

$$\begin{aligned} \int_{T_S - t}^{T_E - t} f_t(\tau)d\tau = \int_{T_S - t}^{T_E - t} \hat{f}_t(\tau)d\tau &+ \big(B_P(T_E - t) - B_P(T_S - t)\big)X_t \\ &+ \big(\tilde{B}_P(T_E - t) - \tilde{B}_P(T_S - t)\big)Y_t. \end{aligned} \tag{54}$$

Since $\int_0^{\Delta} \hat{f}_{T_S}(\tau)\,du = \int_{T_S-t}^{T_E-t} \hat{f}_t(\tau)d\tau$, subtracting (54) from (53), we have

$$P(T_S,T_E)^{-1} = \frac{P(t,T_S)}{P(t,T_E)} e^{B_P(\Delta)X_{T_S} - \left(B_P(T_E-t)-B_P(T_S-t)\right)X_t + \tilde{B}_P(\Delta)Y_{T_S} - \left(\tilde{B}_P(T_E-t)-\tilde{B}_P(T_S-t)\right)Y_t}.$$

Conditioning it on $\mathcal{F}_t$, proves Eq. (52). To prove (51), we consider first the case $t \le T_S$. Now we use the relationship

$$\int_{T_S}^{T_E} r(u)\,du = \int_{T_S}^{T_E} f_u(0)\,du = \int_{T_S}^{T_E} \hat{f}_u(0)\,du + B(0)\int_{T_S}^{T_E} X_u\,du + \tilde{B}(0)\int_{T_S}^{T_E} Y_u\,du. \tag{55}$$

Similarly, subtracting (54) from (55), we have

$$e^{\int_{T_S}^{T_E} r(u)\,du} = \frac{P(t,T_S)}{P(t,T_E)} e^{B(0)\int_{T_S}^{T_E} X_u\,du + \tilde{B}(0)\int_{T_S}^{T_E} Y_u\,du - \left(B_P(T_E-t)-B_P(T_S-t)\right)X_t - \left(\tilde{B}_P(T_E-t)-\tilde{B}_P(T_S-t)\right)Y_t}. \tag{56}$$

For $T_S < t \le T_E$, using (56) applied to $[t, T_E]$, we have

$$e^{\int_{T_S}^{T_E} r(u)\,du} = \frac{e^{\int_{T_S}^{t} r(u)\,du}}{P(t,T_E)} e^{B(0)\int_{t}^{T_E} X_u\,du + \tilde{B}(0)\int_{t}^{T_E} Y_u\,du - \left(B_P(T_E-t)-B_P(0)\right)X_t - \left(\tilde{B}_P(T_E-t)-\tilde{B}_P(0)\right)Y_t}. \tag{57}$$

We can combine (56) and (57) into single expression

$$\begin{aligned} e^{\int_{T_S}^{T_E} r(u)\,du} = {} & \frac{P(t,T_S)}{P(t,T_E)} e^{B(0)\int_{T_S\wedge t}^{T_E} X_u\,du + \tilde{B}(0)\int_{T_S\wedge t}^{T_E} Y_u\,du} \\ & \times e^{-\left(B_P(T_E-T_E\vee t)-B_P(T_S-T_S\vee t)\right)X_t - \left(\tilde{B}_P(T_E-T_E\vee t)-\tilde{B}_P(T_S-T_S\vee t)\right)Y_t}, \end{aligned} \tag{58}$$

which is valid for all $t \le T_E$, where $P(\cdot,\cdot)$ denotes extended zero-coupon bond. Conditioning Eq. (57) on $\mathcal{F}_t$, completes the proof.

□

3.3.1 Gaussian model

First, we consider analytically tractable case of dynamics (9) with deterministic volatility $\sigma_t \equiv 1$ so that the covariance of factors is deterministic and fully specified by $\mathbf{C}(t)$. Kirikos and Novak (1997) and Henrard (2005) study convexity adjustment for Eurodollar futures using Hull–White and single-factor Gaussian HJM model, respectively, Jacquier and Oumgari (2023) consider Gaussian Volterra process and Skov and Skovmand (2021) apply arbitrage-free Nelson–Siegel model for rates dynamics. In Theorem 3.3 we give analytic representation of futures convexity adjustment within Gaussian subclass of Factor HJM model for arbitrary basis $B(\tau)$. This result is a unification of results in the aforementioned works.

Theorem 3.3 *Assume that σ_t in (9) is deterministic. Then convexity adjustments $c^{\mathrm{ED}}(t)$, $c^{\mathrm{ON}}(t)$ in Eqs. (52) and (51) for $t \in [0, T_S]$ are given, respectively, by*

$$
\begin{aligned}
c^{\mathrm{ED}}(t) &= e^{\int_t^{T_S} B_P(T_E-s)^\top \left(B_P(T_E-s)-B_P(T_S-s)\right)\mathbf{C}(s)^\top \mathbf{C}(s)\,ds}, \\
c^{\mathrm{ON}}(t) &= e^{\int_{T_S}^{T_E} B_P(T_E-s)^\top B_P(T_E-t)\mathbf{C}(t)^\top \mathbf{C}(t)\,dt} \times c^{\mathrm{ED}}(t).
\end{aligned} \tag{59}
$$

Proof See Appendix 4.1. □

We highlight that expression (59) can be integrated analytically, if volatility matrix $\mathbf{C}(\cdot)$ is constant. In general, when $\mathbf{C}(\cdot)$ is time-dependent, formula (59) can be calculated by numerical integration. Therefore, evaluation of futures convexity adjustment in Gaussian FHJM is analytic.

3.3.2 CIR SV driver

Let us present analytical solution for $c^{\mathrm{ON}}(t)$ and $c^{\mathrm{ED}}(t)$ in FHJM–SV model with CIR SV driver. Although their settlement values are different, with first based on continuously compounded SOFR rate and second on simple forward rate, we show that both convexity adjustments can be solved using same approach.[3]

Theorem 3.4 *The function $c^{\mathrm{ON}}(t)$ in (51) under CIR SV dynamics (11) for $t \in [0, T_E]$ is given by*

$$
c^{\mathrm{ON}}(t) = c^{\mathrm{ON}}(t;V) = \exp\left[h_0(t) + h_1(t)V\right], \tag{60}
$$

where h_0, h_1 satisfy the following quadratic differential system as a function of $\tau = T_E - t$:

$$
\begin{aligned}
\frac{d}{d\tau}B_1 &= B_1^\top \mathbf{D} + \mathbb{1}_{\tau\le\Delta} B(0), \\
\frac{d}{d\tau}B_2 &= B_2^\top \widetilde{\mathbf{D}} + \mathbb{1}_{\tau\le\Delta} \tilde{B}(0), \\
\frac{d}{d\tau}h_0 &= \kappa\theta h_1, \\
\frac{d}{d\tau}h_1 &= \frac{1}{2}B_1^\top \mathbf{M} B_1 + h_1 B_1^\top \mathbf{C}\beta + B_2^\top \tilde{\Omega} - \kappa h_1 + \frac{1}{2}\vartheta^2 (h_1)^2
\end{aligned} \tag{61}
$$

with the initial condition $B_1(0) = B_2(0) = 0$, $h_0(0) = h_1(0) = 0$.

Here $\vartheta(t)$ is the total volatility of variance defined by

$$
\vartheta(t) = \sqrt{\|\vec{\beta}(t)\|^2 + (\beta_0(t))^2}.
$$

Proof See Appendix 4.2. □

Corollary 3.5 *The function $c^{\mathrm{ED}}(t)$ in (52) under CIR SV dynamics (11) $t \in [0, T_S]$ is given by*

[3] At the same time, convexity adjustment for SOFR futures has broader practical applications since in USD market SOFR futures have successfully replaced Eurodollar futures.

$$c^{\mathrm{ED}}(t) = c^{\mathrm{ED}}(t;V) = \exp\left[h_0(t) + h_1(t)V\right], \tag{62}$$

where h_0, h_1 *satisfy the following quadratic differential system as a function of* $\tau = T_S - t$:

$$\begin{aligned}
\frac{d}{d\tau}B_1 &= B_1^\top \mathbf{D},\\
\frac{d}{d\tau}B_2 &= B_2^\top \tilde{\mathbf{D}},\\
\frac{d}{d\tau}h_0 &= \kappa\theta h_1,\\
\frac{d}{d\tau}h_1 &= \frac{1}{2}B_1^\top \mathbf{M} B_1 + h_1 B_1^\top \mathbf{C}\beta + B_2^\top \tilde{\Omega} - \kappa h_1 + \frac{1}{2}\vartheta^2 (h_1)^2,
\end{aligned} \tag{63}$$

with the initial condition $B_1(0) = B_P(\Delta)$, $B_2(0) = \tilde{B}_P(\Delta)$, $h_0(0) = h_1(0) = 0$.

Convexity adjustments for SOFR and ED futures solve, at the first glance, similar problems (cf. the right-hand side of (61) and (63)). The difference, however, is important and stems from the settlement mechanism of the contracts: SOFR futures settles at T_E into SOFR rate, compounded over $[T_S, T_E]$, whereas Eurodollar futures settlement value is known at T_S. Accruing nature of setting-in-arrears SOFR rate explains the presence of extra terms in the right-hand side of (61).

We emphasize that solutions for SOFR and Eurodollar futures convexities in (60) and (62) are *exact*, not approximate. Next, for valuation of options on rate futures we need the underlying dynamics under forward measure $\mathbb{Q}^{T_E}$, where T_E is option payment date.

Corollary 3.6 *Let* $t \in [0, T_S]$. *The dynamics of* F_t^{ON} *in model* (9) *under* T_E*-forward measure* $\mathbb{Q}^{T_E}$ *is*

$$\begin{aligned}
dF_t^{\mathrm{ON}} &= \left(F_t^{\mathrm{ON}} + \Delta^{-1}\right)\left[-V_t a_0(t)^\top \eta(t)\,dt + \sqrt{V_t}\,a_0(t)^\top\,dW_t^{T_E,(0)} + a_1(t)\sqrt{V_t}\,dw_t^{T_E,(1)}\right],\\
dV_t &= \left[\kappa\theta - \left(\kappa + \beta(t)^\top \eta(t)\right)V_t\right]dt + \sqrt{V_t}\left(\vec{\beta}(t)^\top\,dW_t^{T_E,(0)} + \beta_0(t)\,dw_t^{T_E,(1)}\right),
\end{aligned} \tag{64}$$

where

$$\begin{aligned}
\eta(t) &:= \mathbf{C}(t)^\top B_P(T_E - t)^\top,\\
a_0(t) &:= (B_P(T_E - t) - B_P(T_S - t))\mathbf{C}(t) + h_1(t)\vec{\beta}(t)^\top,\\
a_1(t) &:= \beta_0(t)h_1(t),
\end{aligned} \tag{65}$$

and h_0, h_1 *solve Eq.* (61).

As its proof is similar to that of Theorem 3.6, we omit it. Similarly, we skip the result concerning its $\mathbb{Q}$-dynamics for brevity. We also skip the discussion of moment explosion under the CIR SV driver since this subject is well-covered in the existing

literature, see Andersen and Piterbarg (2007), Glasserman and Kim (2010) and references therein.

3.3.3 Log-normal SV driver

Next, we derive analytic solution for the convexity adjustment function $c^{\mathrm{ON}}(t)$ in Eq. (51) for the log-normal SV driver. We present the exact details in Appendix 4, while here we apply the main result stating that under log-normal SV driver the function $c^{\mathrm{ON}}(t)$ can be decomposed into leading term $E^{[0]}$ and reminder term $R^{[0]}$, where $E^{[0]}$ is exponential-affine function of mean-adjusted volatility $\sigma_t - \theta$.

Theorem 3.5 *Assuming the solution for the convexity adjustment function using leading term $E^{[0]}$ in Eq. (148), we consider futures rate F_t^{ON}, $t \in [0, T_E]$ defined by*

$$\begin{aligned} F_t^{\mathrm{ON}} &= F^{\mathrm{ON}}(t;T_S,T_E) := \frac{1}{\Delta}\left(\frac{P(t,T_S)}{P(t,T_E)}c^{\mathrm{ON}}(t) - 1\right), \\ c^{\mathrm{ON}}(t) &= c^{\mathrm{ON}}(t;\sigma) = \exp\left[h_0(t) + h_1(t)(\sigma - \theta)\right], \end{aligned} \tag{66}$$

where h_0, h_1 satisfy the following quadratic differential system as a function of $\tau = T_E - t$:

$$\begin{aligned} \frac{d}{d\tau}B_1 &= B_1^{\top}\mathbf{D} + \mathbb{1}_{\tau\le\Delta}B(0), \\ \frac{d}{d\tau}B_2 &= B_2^{\top}\tilde{\mathbf{D}} + \mathbb{1}_{\tau\le\Delta}\tilde{B}(0), \\ \frac{d}{d\tau}h_0 &= \theta^2\left(\frac{1}{2}B_1^{\top}\mathbf{M}B_1 + B_2^{\top}\tilde{\Omega}\right) + \kappa_0 h_1 + \frac{1}{2}\vartheta^2\theta^2 h_1^2 + \theta^2 h_1 \cdot B_1^{\top}\mathbf{C}\beta, \\ \frac{d}{d\tau}h_1 &= 2\theta\left(\frac{1}{2}B_1^{\top}\mathbf{M}B_1 + B_2^{\top}\tilde{\Omega}\right) - \kappa_1 h_1 + \vartheta^2\theta h_1^2 + 2\theta h_1 \cdot B_1^{\top}\mathbf{C}\beta, \end{aligned} \tag{67}$$

with the initial condition $B_1(0) = B_2(0) = 0$, $h_0(0) = h_1(0) = 0$.

Here $\vartheta(t)$ is the total volatility of volatility defined by

$$\vartheta(t) = \sqrt{\|\vec{\beta}(t)\|^2 + (\beta_0(t))^2}. \tag{68}$$

Proof See Appendix 4.3. □

In Fig. 1, we display the magnitude of futures convexity adjustment in the FHJM–SV model, calculated using solution of the zeroth-order affine expansion (148) and exact solution, estimated using MC simulations in model dynamics (9) using Nelson–Siegel basis. We use 200,000 trials and daily time steps.

If one needs to estimate only futures convexity adjustment, then extension of affine expansion to first- and second-order is possible.

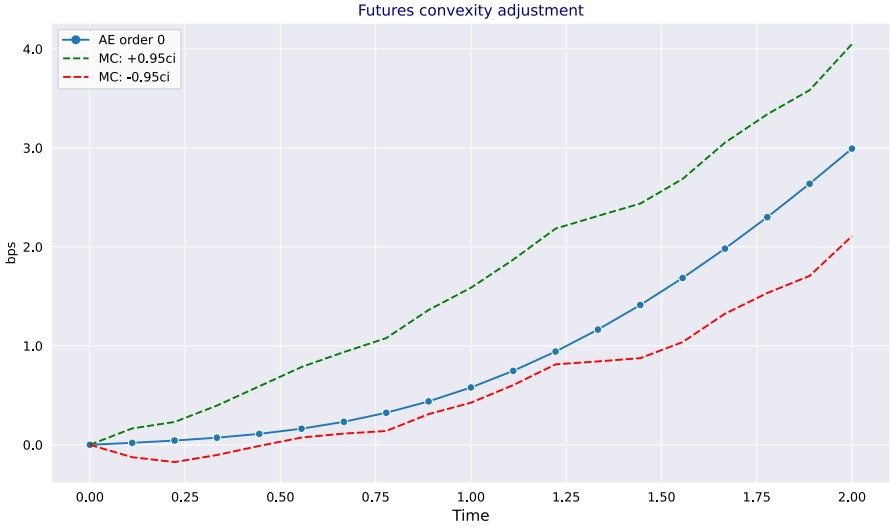


Fig. 1 Futures convexity adjustment computed using Eq. (66) for the expiries up to $2y$. Monte-Carlo simulations are performed using model dynamics (9) with log-normal SV driver (12). Dashed lines labeled by $MC - 0.95ci$ and $MC + 0.95ci$ are MC 95% confidence intervals computed using MC simulations. Solid blue line represents convexity-adjustment calculated using affine expansion in Theorem 3.5. Model parameters are calibrated to SOFR options slice with 103d expiry. Market data is the same as in Sect. 7.7

Corollary 3.7 *The function* $c^{\text{ED}}(t)$ *in* (52) *under CIR SV dynamics* (11) *for* $t \in [0, T_S]$ *is given by*

$$c^{\text{ED}}(t) = c^{\text{ED}}(t;\sigma) = \exp\left[h_0(t) + h_1(t)(\sigma - \theta)\right], \tag{69}$$

where h_0, h_1 *satisfy the following quadratic differential system as a function of* $\tau = T_S - t$:

$$\begin{aligned}
\frac{d}{d\tau}B_1 &= B_1^\top \mathbf{D},\\
\frac{d}{d\tau}B_2 &= B_2^\top \tilde{\mathbf{D}},\\
\frac{d}{d\tau}h_0 &= \theta^2\Big(\frac{1}{2}B_1^\top \mathbf{M} B_1 + B_2^\top \tilde{\Omega}\Big) + \kappa_0 h_1 + \frac{1}{2}\vartheta^2\theta^2 h_1^2 + \theta^2 h_1 \cdot B_1^\top \mathbf{C}\beta,\\
\frac{d}{d\tau}h_1 &= 2\theta\Big(\frac{1}{2}B_1^\top \mathbf{M} B_1 + B_2^\top \tilde{\Omega}\Big) - \kappa_1 h_1 + \vartheta^2\theta h_1^2 + 2\theta h_1 \cdot B_1^\top \mathbf{C}\beta,
\end{aligned} \tag{70}$$

with the initial condition $B_1(0) = B_P(\Delta)$, $B_2(0) = \tilde{B}_P(\Delta)$, $h_0(0) = h_1(0) = 0$.

Convexity adjustment function in (66) and (69) does not include quadratic term because the presence of higher-order terms hampers the construction of analytically tractable model for rate futures. In practice, we observe that higher order terms usually provide insignificant contribution across the wide range of model parameters as they are very small compared to first two terms. With the linear

exp-affine function for $c^{\mathrm{ON}}(t)$, we obtain analytically tractable dynamics of SOFR futures.

Corollary 3.8 *Let $t \le T_S$. In model (9), the risk-neutral dynamics of F_t^{ON}, defined by (66), is given by*

$$\begin{aligned} dF_t^{\mathrm{ON}} &= \left(F_t^{\mathrm{ON}} + \Delta^{-1}\right)\sigma_t\left(a_0(t)\,dW_t^{\mathbb{Q},(0)} + a_1(t)\,dw_t^{\mathbb{Q},(1)}\right), \quad F_0 = F \\ d\sigma_t &= \left(\kappa_1 + \kappa_2\sigma_t\right)(\theta - \sigma_t)dt + \sigma_t\vec{\beta}(t)^{\top}dW_t^{\mathbb{Q},(0)} + \beta_0(t)\sigma_t dw_t^{\mathbb{Q},(1)}, \quad \sigma_0 = \sigma \end{aligned} \tag{71}$$

where $\tau_S = T_S - t$, $\tau_E = T_E - t$ and $a_0(t)$, $a_1(t)$ are defined by

$$\begin{aligned} a_0(t) &:= (B_P(\tau_E) - B_P(\tau_S))\mathbf{C}(t) + h_1(t)\vec{\beta}(t)^{\top}, \\ a_1(t) &:= \beta_0(t)h_1(t), \end{aligned} \tag{72}$$

and h_0, h_1 solve Eq. (67).

Proof Due to bond-reconstruction formula in FHJM–SV model, we have

$$\begin{aligned} \frac{P(t,T_S)}{P(t,T_E)} &= \frac{P(0,T_S)}{P(0,t)}\exp\left(-B_P(\tau_S)X_t - \tilde{B}_P(\tau_S)Y_t\right)\left[\frac{P(0,T_E)}{P(0,t)}\exp\left(-B_P(\tau_E)X_t - \tilde{B}_P(\tau_E)Y_t\right)\right]^{-1} = \\ &= \frac{P(0,T_S)}{P(0,T_E)}\exp\left[(B_P(\tau_E) - B_P(\tau_S))X_t + (\tilde{B}_P(\tau_E) - \tilde{B}_P(\tau_S))Y_t\right]. \end{aligned}$$

Thus, we obtain

$$\begin{aligned} d\left[\frac{P(t,T_S)}{P(t,T_E)}\right] &= \frac{P(t,T_S)}{P(t,T_E)}\left[(B_P(\tau_E) - B_P(\tau_S))\,dX_t + (\ldots)\,dt\right] \\ &= \frac{P(t,T_S)}{P(t,T_E)}\left[\sigma_t(B_P(\tau_E) - B_P(\tau_S))\mathbf{C}(t)\,dW_t^{(0)} + (\ldots)\,dt\right]. \end{aligned}$$

Next, since

$$dc^{\mathrm{ON}}(t) = \sigma_t c^{\mathrm{ON}}(t)\left[h_1(t)\left(\vec{\beta}(t)^{\top}\,dW_t^{\mathbb{Q},(0)} + \beta_0(t)\,dw_t^{\mathbb{Q},(1)}\right) + (\ldots)dt\right]$$

it follows from definition (66) that

$$\begin{aligned} dF_t^{\mathrm{ON}} = \frac{1}{\Delta}d\left[\frac{P(t,T_S)}{P(t,T_E)}c^{\mathrm{ON}}(t)\right] &= \left(F_t^{\mathrm{ON}} + \Delta^{-1}\right)\sigma_t\Big[(B_P(\tau_E) - B_P(\tau_S))\mathbf{C}(t)\,dW_t^{\mathbb{Q},(0)} \\ &+ h_1(t)\left(\vec{\beta}(t)^{\top}\,dW_t^{\mathbb{Q},(0)} + \beta_0(t)\,dw_t^{\mathbb{Q},(1)}\right) + (\ldots)dt\Big]. \end{aligned}$$

By noting that futures rate has zero-drift under $\mathbb{Q}$, see Eq. (48), we complete the proof. □

Remark. From Theorem 3.8 we see that futures rate follows displaced log-normal dynamics in model (9).

In Sect. 4 we develop the framework for analytical valuation of options on rate futures. Hence, we need the underlying dynamics in T_E-forward measure $\mathbb{Q}^{T_E}$.

Theorem 3.6 *Let $t \le T_S$. The dynamics of F_t^{ON} in model (9) under T_E-forward measure $\mathbb{Q}^{T_E}$ is*

$$\begin{aligned} dF_t^{\text{ON}} &= \left(F_t^{\text{ON}} + \Delta^{-1}\right)\left[-\sigma_t^2 a_0(t)^\top \eta(t)\,dt + \sigma_t a_0(t)^\top\, dW_t^{T_E,(0)} + a_1(t)\sigma_t\, dw_t^{T_E,(1)}\right], \\ d\sigma_t &= [\kappa_1\theta - (\kappa_1 - \kappa_2\theta)\sigma_t - (\kappa_2 + \vec{\beta}(t)^\top \eta(t))\sigma_t^2]\,dt + \sigma_t\vec{\beta}(t)^\top\, dW_t^{T_E,(0)} + \beta_0(t)\sigma_t dw_t^{T_E,(1)}. \end{aligned} \tag{73}$$

where

$$\eta(t) := \mathbf{C}(t)^\top B_P(T_E - t)^\top. \tag{74}$$

Here we must impose the condition

$$\kappa_2 > \vec{\beta}(t)^\top \mathbf{C}^\top(t) B_P(T_E - t). \tag{75}$$

so that the volatility process σ_t in Eq. (73) admits strong solution and to ensure equivalence of measures $\mathbb{Q} \sim \mathbb{Q}^{T_E}$, which is required for the application of Girsanov theorem in Theorem 3.6 [see Sepp and Rakhmonov (2023b) for details].

3.3.4 Martingale property of futures prices

Finally, we study the problem of loss of martingale property in futures prices using measure-change argument and utilize the idea that futures price F_t in approximate dynamics (71) is a true martingale if auxiliary volatility process, as defined in (156), does not explode. This idea has been successfully applied in Sin (1998), Lewis (2000), Lucic (2004), Lions and Musiela (2007), among others, although in a different contexts. We find conditions under which futures price F_t^{ON} preserve the martingale property over a given time horizon $t \in [0, T^*]$, thus ensuring that model for futures rate is well-specified.

Theorem 3.7 *Consider approximate futures dynamics F_t^{ON} given by (71) over time horizon $t \in [0, T^*]$. Assume that $\kappa_1 \ge 0$, $\theta > 0$. Then, the futures price process is a true martingale for $t \in [0, T^*]$ if*

$$\kappa_2 > \max_{t\in[0,T^*]}\left(0, \vec{\beta}(t)^\top a_0(t) + \beta_0(t)a_1(t)\right), \tag{76}$$

where $a_0(t)$, $a_1(t)$ are defined in Eq. (72).

Proof is given in Appendix 4.4.

Remark. Using exactly same approach, we can derive same conditions under which Eurodollar futures price F_t^{ED} preserve the martingale property over some finite horizon. We skip its proof for sake of brevity.

4 Valuation of rates derivatives using MGF

In this section, we derive analytic formulas for valuation of swaptions and options of rate futures under generic FHJM–SV framework using moment generating function (MGF) corresponding to a specific FHJM–SV model. We emphasize that these valuation formulas can be applied to any concrete FHJM–SV model with analytic MGF. In the next section, we derive solutions for MGFs in the FHJM–SV model with CIR and log-normal SV driver.

4.1 Swaptions

We consider vanilla call and put payoffs, denoted by $u^{call}(s_T)$ and $u^{put}(s_T)$ respectively, on underlying swap rate s_T settled at time T

$$u^{call}(s_T) = A(T)\max(s_T - K, 0),$$
$$u^{put}(s_T) = A(T)\max(K - s_T, 0) = A(T)\big[\max(s_T - K, 0) - (s_T - K)\big],$$

where $A(T)$ is the annuity. By risk-neutral approach, the value function of the option with payoff $u(s_T)$, denoted by $U(\tau, T; s, v)$, is valued, using the annuity as the numeraire, as follows

$$U(\tau, T; s, v) = A(t)\mathbb{E}^A\left[u(s_T)\middle| s_t = s, v_t = v\right], \tag{77}$$

where v_t stands for either the volatility or variance variable. Assuming that MFG is available analytically, we then adapt Lewis–Lipton approach for valuation of vanilla options using Fourier transform, see Lewis (2000) and Lipton (2001).

Definition 4.1 (*Moment generating function*) Under the annuity measure, the MGF of the swap rate s_T in dynamics (39) is defined by

$$G(\tau, T, s, v; \phi) = \mathbb{E}^A\left[e^{-\phi s_T}\middle| s_t = s, v_t = v\right]. \tag{78}$$

We assume that G exists. The existence must be analyzed case-by-case for specific choices of FHJM–SV models. The analysis for the FHJM–SV model with the log-normal volatility follows Theorem 4.2 in Sepp and Rakhmonov (2023b) and hinges upon the martingale property of $\frac{A(t)/A(0)}{N(t)}$, which is defined in (137). To ensure it, we gave sufficient condition in Eq. (33), see Corollary 3.1 and its proof in Appendix 3.1.

Corollary 4.1 (PDF) *The probability density function* (*PDF*) *of* s_T, *denoted by* $p(\tau, T, s, v; s')$ *can be computed by the Fourier inversion*:

$$p(\tau, T, s, v; s') = \frac{1}{\pi}\Re\left[\int_0^\infty \exp\left\{\phi s'\right\} G(\tau, T, s, v; \phi)\, d\phi\right], \tag{79}$$

where $\phi = \phi_r + ip$, $i = \sqrt{-1}$, *and* $d\phi \equiv dp$.

Proposition 4.1 *We assume that MGF* $G(\tau, T, s, v;\phi)$ *is the MGF of the swap rate* s_T *with transform variable* ϕ *defined in* (78). *The value of call option in* (77) *is given by*

$$U(\tau, T;s, v) = \frac{1}{\pi}A(t)\Re\left[\int_0^\infty \frac{e^{\left(-\frac{1}{2}+ip\right)K}}{\left(-\frac{1}{2}+ip\right)^2}G(\tau, T, s, v;\phi = -\frac{1}{2}+ip)dp\right], \quad (80)$$

where $p \in \mathbb{R}$.

Proof Using Eq. (77) along with Eq. (79), we obtain

$$U(\tau, T;s, v) = A(t)\Re\left[\int_{-\infty}^\infty \max\left\{s' - K, 0\right\}\left[\frac{1}{\pi}\int_0^\infty e^{\phi s'}G(\tau, T, s, v;\phi)d\phi\right]ds'\right]. \quad (81)$$

Assuming that the inner integrals are finite, we exchange the integration order and the valuation formula becomes

$$U(\tau, T;s, v) = \frac{1}{\pi}A(t)\Re\left[\int_0^\infty \widehat{u}(\phi)G(\tau, T, s, v;\phi)d\phi\right], \quad (82)$$

where $\widehat{u}(\phi)$ is the transformed payoff function:

$$\begin{aligned}\widehat{u}(\phi) &= \int_{-\infty}^\infty e^{\phi s'}\max\left\{s' - K, 0\right\}ds' = \int_K^\infty e^{\phi s'}(s' - K)ds' \\ &= \frac{1}{\phi}\int_K^\infty (s' - K)e^{\phi s'}ds' = \frac{e^{\phi K}}{\phi}\int_K^\infty (s' - K)e^{\phi(s'-K)}ds' = \frac{e^{\phi K}}{\phi}\int_0^\infty e^{\phi u}du = \frac{e^{\phi K}}{\phi^2},\end{aligned} \quad (83)$$

provided $\Re[\phi] < 0$. Setting $\phi = -\frac{1}{2} + ip$, we obtain valuation formula (80).

□

Corollary 4.2 (*Swaption valuation*) *The values of call option* (*payer swaption*) *and put option* (*receiver swaption*) *on swap rates are computed by*

$$\begin{aligned}U^{call}(\tau, T, s) &= U(\tau, T;s, v), \\ U^{put}(\tau, T, s) &= U(\tau, T;s, v) - A(t)(s - K),\end{aligned} \quad (84)$$

with $U(\tau, T;s, v)$ *defined in Eq.* (80).

4.2 Options on rates futures

We consider the futures rate and introduce log-shifted futures rate ζ_t defined as

$$\zeta_t = \ln\left(F_t + \Delta^{-1}\right), \quad t \geq 0. \tag{85}$$

We now assume that $G(\tau, T, \zeta, v;\phi)$ is the MGF corresponding to ζ_T defined by analogy to Eq. (78). Let us consider call and put option on the futures rate with the following payoff settled at option expiry T:

$$\begin{aligned} c(F_T, K) &= \max(F_T - K, 0) = F_T - \min(F_T, K) = F_T - \min(e^{\zeta_T} - \Delta^{-1}, K) \\ &= F_T + \Delta^{-1} - \min(e^{\zeta_T}, K + \Delta^{-1}), \\ p(F_T, K) &= \max(K - F_T, 0) = K - \min(F_T, K) = K - \min(e^{\zeta_T} - \Delta^{-1}, K) \\ &= K + \Delta^{-1} - \min(e^{\zeta_T}, K + \Delta^{-1}). \end{aligned} \tag{86}$$

We note that options on futures rate (for instance, SOFR futures options traded on CME) are usually American-style and do not have futures-style margining. We treat them as European and ignore their early exercise premium, which is usually insignificant.

For notational simplicity, we denote option payment date by T, i.e. $T := T_E$. Thus, we consider the valuation of capped payoff under T-forward measure $\mathbb{Q}^T$:

$$U(\tau, T;\zeta, v) = P(t, T)\mathbb{E}^{\mathbb{Q}^T}\left[\min(e^{\zeta_T}, K + \Delta^{-1})\middle|\zeta_t = \zeta, v_t = v\right]. \tag{87}$$

We similarly apply Lewis-Lipton approach for valuation of vanilla options using Fourier transform.

Proposition 4.2 *The valuation formula for the capped payoff is given by*

$$U(\tau, T;\zeta, v) = P(t, T)\frac{K + \Delta^{-1}}{\pi}\Re\left[\int_0^\infty e^{(-\frac{1}{2}+ip)\zeta^*}\frac{1}{p^2 + 1/4}G(\tau, T, \zeta, v;\phi = -\frac{1}{2} + ip)dp\right], \tag{88}$$

where $\zeta^* = \ln\left(\frac{K+\Delta^{-1}}{F+\Delta^{-1}}\right)$.

Proof Using the MGF, the density of ζ_τ is computed by the Fourier inversion

$$p(\tau, T, \zeta, v;\zeta') = \frac{1}{\pi}\Re\left[\int_0^\infty \exp\left\{\phi\zeta'\right\}G(\tau, T, \zeta, v;\phi)\, d\phi\right]. \tag{89}$$

We have

$$U(\tau, T;\zeta, v) = P(t, T)\Re\left[\int_{-\infty}^\infty \min\left\{e^{\zeta'}, K + \Delta^{-1}\right\}\left[\frac{1}{\pi}\int_0^\infty e^{\phi\zeta'}G(\tau, T, \phi)d\phi\right]d\zeta'\right].$$

Assuming that the inner integrals are finite, we exchange the integration order to obtain

$$U(\tau, T, \zeta) = \frac{1}{\pi}P(t, T)\Re\left[\int_0^\infty \widehat{u}(\phi)G(\tau, T, \phi)d\phi\right]. \tag{90}$$

Here $\widehat{u}(\phi)$ is the transformed payoff function

$$\begin{aligned}\widehat{u}(\phi) &= \int_{-\infty}^{\infty} e^{\phi\zeta'} \min\left\{e^{\zeta'}, K+\Delta^{-1}\right\} d\zeta' = \frac{e^{(\phi+1)k^*}}{\phi+1} - e^{k^*}\frac{1}{\phi}e^{\phi\zeta^*} = -e^{(\phi+1)\zeta^*}\frac{1}{(\phi+1)\phi} \\ &= -e^{\zeta^*}e^{\phi\zeta^*}\frac{1}{(\phi+1)\phi} = -(K+\Delta^{-1})e^{\phi\zeta^*}\frac{1}{(\phi+1)\phi},\end{aligned} \tag{91}$$

where $\zeta^* = \ln\left(\frac{K+\Delta^{-1}}{F+\Delta^{-1}}\right)$, $k^* = \ln(K+\Delta^{-1})$ with the first integral being finite for $\Re[\phi] > -1$ and the second integral being finite for $\Re[\phi] < 0$. Thus, integral (91) is finite for $-1 < \Re[\phi] < 0$. Setting $\phi = -1/2 + ip$, we derive Eq. (88).

□

Corollary 4.3 (*Valuation of options on rate futures*) *The value of call and put option on rate future is computed, respectively, as*

$$\begin{aligned}U^{call}(\tau,T,\zeta) &= P(t,T)(R_t+\Delta^{-1}) - U(\tau,T,\zeta), \\ U^{put}(\tau,T,\zeta) &= P(t,T)(K+\Delta^{-1}) - U(\tau,T,\zeta),\end{aligned} \tag{92}$$

with $U(\tau,T,\zeta)$ *defined in*(88) *and* $R_t = \frac{1}{\Delta}\left(\frac{P(t,T_S)}{P(t,T_E)} - 1\right)$

As we ignored early exercise feature of options on rate futures, the formula for call option in (92) can be deduced from put-call parity:

$$U^{call}(\tau,T,\zeta) - U^{put}(\tau,T,\zeta) = P(t,T)\left(\mathbb{E}^T\left[F_T\middle|\mathcal{F}_t\right] - K\right) = P(t,T)\left(R_t - K\right), \tag{93}$$

where we used the fact that [see Lyashenko and Mercurio (2019)]

$$\mathbb{E}^T\left[F_T\middle|\mathcal{F}_t\right] \equiv \mathbb{E}^T\left[F^{\text{ON}}(T_S,T_E)\middle|\mathcal{F}_t\right] = \frac{1}{\Delta}\left(\frac{P(t,T_S)}{P(t,T_E)} - 1\right).$$

5 Solution to MGF for FHJM–SV model with CIR SV

We first derive the analytic solution for the MGF of the state variable z_t in FHJM model with CIR SV dynamics. To keep notations consistent, we introduce mean-adjusted process v_t defined by

$$v_t = V_t - \theta, \tag{94}$$

where V_t is the CIR SV process with the dynamic in Eq. (11).

We further introduce generic state variable process z_t which represents either the swap rate s_t in Eq. (41) or the log-shifted futures price ζ_t in Eq. (85) with futures dynamics given by Eq. (64). We assume that under the pricing measure $\hat{\mathbb{Q}}$ the dynamics of state variables $\{z_t, v_t\}$ are given by

$$
\begin{aligned}
dz_t &= -\hat{b}(t)(v_t+\theta)\,dt + \sqrt{v_t+\theta}\left[\hat{a}_0(t)^\top\, d\hat{W}_t^{(0)} + \hat{a}_1(t)\,d\hat{w}_t^{(1)}\right],\\
dv_t &= (\hat{\kappa}_0(t) - \hat{\kappa}_1(t)v_t)dt + \sqrt{v_t+\theta}\left[\vec{\beta}(t)^\top\, d\hat{W}_t^{(0)} + \beta_0(t)\,d\hat{w}_t^{(1)}\right],
\end{aligned}
\tag{95}
$$

where $\hat{a}_0(t)$ and $\hat{a}_1(t)$ are, respectively, vector and scalar functions, assumed to be deterministic function of time.

Theorem 5.1 *We consider moment-generating function (MGF) for the variable z_T, denoted by $G(\tau,T,z,v;\phi)$, using transform variable $\phi\in\mathbb{C}$. Then $G(\tau,T,z,v;\phi)$ is given by the exponential-affine form*:

$$
G(\tau,T,z,v;\phi) = \exp\left\{-\phi z + \sum_{k=0}^{1} A^{(k)}(\tau,T;\phi)v^k\right\},
\tag{96}
$$

where vector function $\mathbf{A}(\tau) = \left\{A^{(k)}(\tau,T;\phi)\right\}$, $k=0,1$, solves the following quadratic differential system as a function of τ:

$$
\begin{aligned}
A_\tau^{(k)} &= \mathbf{A}^\top M^{(k)}\mathbf{A} + \left(L^{(k)}\right)^\top \mathbf{A} + H^{(k)},\\
M^{(k)} &= \left\{\begin{pmatrix} 0 & 0\\ 0 & \frac{\theta\vartheta^2}{2}\end{pmatrix}, \begin{pmatrix} 0 & 0\\ 0 & \frac{\vartheta^2}{2}\end{pmatrix}\right\},\\
L^{(k)} &= \left\{\begin{pmatrix} 0\\ \hat{\kappa}_0 - \theta(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi\end{pmatrix}, \begin{pmatrix} 0\\ -\hat{\kappa}_1 - (\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi\end{pmatrix}\right\},\\
H^{(k)} &= \left\{\frac{1}{2}\theta(\|\hat{a}_0\|^2 + \hat{a}_1^2)\phi^2 + \theta\phi\hat{b},\ \frac{1}{2}(\|\hat{a}_0\|^2 + \hat{a}_1^2)\phi^2 + \phi\hat{b}\right\},
\end{aligned}
\tag{97}
$$

with the initial condition $\mathbf{A}(0) = (0,0)^\top$, where we omit argument $T-\tau$ in functions $\hat{\kappa}_i, \hat{a}_i, \hat{b}, \vec{\beta}, \beta_0, \vartheta$.

We emphasize that although solution (96) is exact, parameters of the SDE (95) are inherently time-dependent, even when parameters of the variance process V_t in Eq. (11) are constant in time. Thus, we always need to resort to numerical solution of the quadratic ODEs (97).

Proof The MGF G in Eq. (96) solves the problem

$$
-\frac{\partial G}{\partial\tau} + \mathcal{L}G = 0, \qquad G(0,T;z,v;\phi) = e^{-\phi z},
\tag{98}
$$

where operator $\mathcal{L}$ is defined by $\mathcal{L} = \mathcal{L}^{(z)} + \mathcal{L}^{(v)}$ with

$$\begin{aligned}\mathcal{L}^{(z)} &= -\hat{b}(T-\tau)(v+\theta)\frac{\partial}{\partial z} + \frac{1}{2}\big(\|\hat{a}_0(T-\tau)\|^2 + \hat{a}_1(T-\tau)^2\big)(v+\theta)\frac{\partial^2}{\partial z^2} \\ &\quad + (v+\theta)\Big[\langle \hat{a}_0(T-\tau), \vec{\beta}(T-\tau)\rangle + \hat{a}_1(T-\tau)\beta_0(T-\tau)\Big]\frac{\partial^2}{\partial z \partial v}, \\ \mathcal{L}^{(v)} &= \big[\hat{\kappa}_0(T-\tau) - \hat{\kappa}_1(T-\tau)v\big]\frac{\partial}{\partial v} + \frac{1}{2}\vartheta(T-\tau)^2(v+\theta)\frac{\partial^2}{\partial v^2}.\end{aligned} \tag{99}$$

Since the problem (98) is affine in v, we take as an ansatz the exponential-affine function in Eq. (96). To obtain the ODEs (97) for $A^{(k)}$, we match the terms with v^k, $k = 0, 1$. Initial conditions for ODEs (97) are set to reproduce initial conditions of the problem (98). □

Corollary 5.1 (*Affine solution for swap rate*) *Moment generating function in Eq.* (96) *for the state-independent swap rate* s_T *in Eq.* (39), *defined by*

$$G(\tau, T, s, v; \phi) = \mathbb{E}^A\Big[e^{-\phi s_T}\Big| s_t = s, v_t = v\Big] \tag{100}$$

can be found as a solution of the problem (97) *by setting*

$$\begin{aligned}&\hat{a}_0(t) = a(t), \quad \hat{a}_1(t) \equiv 0, \quad \hat{b}(t) \equiv 0, \\ &\hat{\kappa}_0(t) := \theta\beta_2(t), \quad \hat{\kappa}_1(t) := \kappa - \beta_2(t)\end{aligned}$$

everywhere in Theorem 5.1, *where* $a(t)$ *and* $\beta_2(t)$ *are defined in Eq.* (40).

Corollary 5.2 (*Affine solution for log-shifted futures price*) *The moment generating function for the log-shifted futures price* ζ_T *in Eq.* (96)

$$G(\tau, T, \zeta, v; \phi) = \mathbb{E}^{\mathbb{Q}^T}\Big[e^{-\phi\zeta_T}\Big| \zeta_t = \zeta, v_t = v\Big] \tag{101}$$

can be found as a solution of the problem (97) *by setting*

$$\begin{aligned}&\hat{a}_0(t) = a_0(t), \quad \hat{a}_1(t) = a_1(t), \\ &\hat{b}(t) = a_0(t)^\top\eta(t) + \frac{1}{2}\big(\|a_0(t)\|^2 + a_1(t)^2\big), \\ &\hat{\kappa}_0(t) := -\theta\vec{\beta}\eta, \quad \hat{\kappa}_1(t) := \kappa + \vec{\beta}\eta\end{aligned}$$

everywhere in Theorem 5.1, *where* $a_0(t), a_1(t), \eta(t)$ *are defined in Eq.* (65).

6 Solution to MGF for FHJM–SV model with log-normal SV driver

We now derive an analytic approximation for the MGF in the FHJM model with log-normal SV dynamics in Eq. (12). We first introduce mean-adjusted volatility process v_t defined by

$$v_t = \sigma_t - \theta.$$

We then introduce generic state variable process z_t which stands for either the swap rate s_t in Eq. (43) or the log-shifted futures price ζ_t in Eq. (85) with futures rate F_t defined in Eq. (73). We assume that under pricing measure $\hat{\mathbb{Q}}$ the dynamics of state variables $\{z_t, v_t\}$ are given by

$$\begin{aligned} dz_t &= -\hat{b}(t)(v_t+\theta)^2\,dt + (v_t+\theta)\Big[\hat{a}_0(t)^\top\,d\hat{W}_t^{(0)} + \hat{a}_1(t)\,d\hat{w}_t^{(1)}\Big], \\ dv_t &= (\hat{\kappa}_0(t) - \hat{\kappa}_1(t)v_t - \hat{\kappa}_2(t)v_t^2)dt + (v_t+\theta)\Big[\vec{\beta}(t)^\top\,d\hat{W}_t^{(0)} + \beta_0(t)\,d\hat{w}_t^{(1)}\Big], \end{aligned} \tag{102}$$

where $\hat{W}^{(0)}$ is d-dimensional vector of independent Brownian motions and $\hat{w}^{(1)}$ is one-dimensional Brownian motion under $\hat{\mathbb{Q}}$. We keep the notations generic so that pricing measure $\hat{\mathbb{Q}}$ could be identified later because it depends on the choice of the instrument. Here $\hat{a}_0(t)$ and $\hat{a}_1(t)$ are, respectively, vector and scalar leverage functions, which are deterministic functions of time.

6.1 Affine expansion of moment generating function

We consider the following valuation problem

$$U(\tau, T; z, v) = \hat{\mathbb{E}}\big[u(z_T)\big|z_t = z, v_t = v\big] = \hat{\mathbb{E}}\big[u(z_T)\big|\mathcal{F}_t\big], \tag{103}$$

where $\tau = T - t$. Using Theorem 4.1 in Sepp and Rakhmonov (2023b), we can show that function U solves the problem (we emphasize that the classic Feynman–Kac formula does not apply directly because of the non-linear drift of the volatility)

$$\begin{aligned} -\frac{\partial U}{\partial \tau} + \mathcal{L}U &= 0, \\ U(0, T; z, v) &= u(z), \end{aligned} \tag{104}$$

where operator $\mathcal{L}$ is defined by $\mathcal{L} = \mathcal{L}^{(z)} + \mathcal{L}^{(v)}$ with

$$\begin{aligned} \mathcal{L}^{(z)} &= -\hat{b}(T-\tau)(v+\theta)^2\frac{\partial}{\partial z} + \frac{1}{2}\big(\|\hat{a}_0(T-\tau)\|^2 + \hat{a}_1(T-\tau)^2\big)(v+\theta)^2\frac{\partial^2}{\partial z^2} \\ &\quad + (v+\theta)^2\Big[\langle\hat{a}_0(T-\tau), \vec{\beta}(T-\tau)\rangle + \hat{a}_1(T-\tau)\beta_0(T-\tau)\Big]\frac{\partial^2}{\partial z \partial v}, \\ \mathcal{L}^{(v)} &= \big[\hat{\kappa}_0(T-\tau) - \hat{\kappa}_1(T-\tau)v - \hat{\kappa}_2(T-\tau)v^2\big]\frac{\partial}{\partial v} + \frac{1}{2}\vartheta(T-\tau)^2(v+\theta)^2\frac{\partial^2}{\partial v^2}. \end{aligned} \tag{105}$$

Here ϑ is defined in Eq. (68).

We introduce the moment-generating function (MGF) for the variable $z(T)$ denoted by $G(\tau, T; z, v)$ using transform variable $\phi \in \mathbb{C}$ as follows

$$G(\tau, T; z, v) = \hat{\mathbb{E}}\Big[e^{-\phi z_T}\Big|z_t = z, v_t = v\Big] = \hat{\mathbb{E}}\Big[e^{-\phi z_T}\Big|\mathcal{F}_t\Big]. \tag{106}$$

6.2 First-order affine expansion

Theorem 6.1 (First-order affine expansion) *The MGF function defined in Eq.* (106) *can be decomposed as follows*:

$$G(\tau,T,z,v;\phi)=\exp\{-\phi z\}\left[E^{[1]}(\tau,T,v;\phi)+R^{[1]}(\tau,T,v;\phi)\right], \tag{107}$$

where $E^{[1]}$ *is the leading term and* $R^{[1]}$ *is the remainder term.*

The leading term $E^{[1]}(\tau,T;z,v)$ *is given by the exponential-affine form*:

$$E^{[1]}(\tau,T,v;\phi)=\exp\left\{\sum_{k=0}^{2}A^{(k)}(\tau,T;\phi)v^k\right\}, \tag{108}$$

where vector function $\mathbf{A}(\tau)=\left\{A^{(k)}(\tau,T;\phi)\right\}$, $k=0,1,2$, *solves the following quadratic differential system as a function of* τ:

$$\begin{aligned}
A^{(k)}_{\tau}&=\mathbf{A}^{\top}M^{(k)}\mathbf{A}+\left(L^{(k)}\right)^{\top}\mathbf{A}+H^{(k)},\\
M^{(k)}&=\left\{\begin{pmatrix}0&0&0\\0&\frac{\theta^2\vartheta^2}{2}&0\\0&0&0\end{pmatrix},\begin{pmatrix}0&0&0\\0&\theta\vartheta^2&\theta^2\vartheta^2\\0&\theta^2\vartheta^2&0\end{pmatrix},\begin{pmatrix}0&0&0\\0&\frac{\vartheta^2}{2}&2\theta\vartheta^2\\0&2\theta\vartheta^2&2\theta^2\vartheta^2\end{pmatrix}\right\},\\
L^{(k)}&=\left\{\begin{pmatrix}0\\\hat{\kappa}_0-\theta^2(\langle\hat{a}_0,\beta\rangle+\beta_0\hat{a}_1)\phi\\\theta^2\vartheta^2\end{pmatrix},\begin{pmatrix}0\\-\hat{\kappa}_1-2\theta(\langle\hat{a}_0,\beta\rangle+\beta_0\hat{a}_1)\phi\\2(\hat{\kappa}_0+\theta\vartheta^2-\theta^2(\langle\hat{a}_0,\beta\rangle+\beta_0\hat{a}_1)\phi)\end{pmatrix},\right.\\
&\qquad\left.\begin{pmatrix}0\\-(\langle\hat{a}_0,\beta\rangle+\beta_0\hat{a}_1)\phi-\hat{\kappa}_2\\\vartheta^2-2\hat{\kappa}_1-4\theta(\langle\hat{a}_0,\beta\rangle+\beta_0\hat{a}_1)\phi\end{pmatrix}\right\},\\
H^{(k)}&=\left\{\frac{1}{2}\theta^2(\|\hat{a}_0\|^2+\hat{a}_1^2)\phi^2+\theta^2\phi\hat{b},\ \theta(\|\hat{a}_0\|^2+\hat{a}_1^2)\phi^2+2\theta\phi\hat{b},\ \frac{1}{2}(\|\hat{a}_0\|^2+\hat{a}_1^2)\phi^2+2\phi\hat{b}\right\},
\end{aligned} \tag{109}$$

with the initial condition $\mathbf{A}(0)=(0,0,0)^{\top}$, *where we omit argument* $T-\tau$ *in functions* $\hat{\kappa}_i,\hat{a}_i,\hat{b},\vec{\beta},\beta_0,\vartheta$.

The remainder term $R^{[1]}(\tau,T,v;\phi)$ *solves the following problem* (*we omit arguments for brevity*):

$$\begin{aligned}
-R^{[1]}_{\tau}+\mathcal{L}^{(v)}R^{[1]}&=-F^{[1]}(v,A^{(1)},A^{(2)})E^{[1]}(\tau,T,v;\phi)\\
R^{[1]}(0,T,v;\phi)&=0,
\end{aligned} \tag{110}$$

with operator $\mathcal{L}^{(v)}$ *defined in Eq.* (105) *and residual function* $F^{[1]}$ *given by*

$$
\begin{aligned}
&F^{[1]}(v, A^{(1)}, A^{(2)}) = \sum_{n=3}^{4} C^{(n)}(\tau; A^{(1)}, A^{(2)}) v^n, \\
&C^{(3)}(\tau) = 2A^{(2)}\left(\vartheta^2\left(2\theta A^{(2)} + A^{(1)}\right) - (\langle \hat{a}_0, \beta\rangle + \beta_0 \hat{a}_1)\phi - \hat{\kappa}_2\right), \\
&C^{(4)}(\tau) = 2\vartheta^2 (A^{(2)})^2.
\end{aligned} \tag{111}
$$

Proof It follows by direct substitution of (107) into Eq. (106) and collecting the terms.

□

Corollary 6.1 (*Estimate of the first-order remainder term*). *We obtain the following estimate for the remainder term* $R^{[1]}$ *in Eq.* (110)

$$
\left| R^{[1]}(\tau, T, v; \phi) \right| \leq \sum_{n=3}^{4} C^{(4)}(\tau^*) \times M_{\sigma}^{(n)}(\tau^*), \tag{112}
$$

where $M_{\sigma}^{(n)}(\tau)$ *is the nth central moment of the volatility process defined by*:

$$
M_{\sigma}^{(n)}(\tau) \equiv \hat{\mathbb{E}}\left[(\sigma(\tau) - \theta)^n\right] = \mathbb{E}\left[v^n(\tau)\right]. \tag{113}
$$

Proof By analogy to the proof of Corollary 4.3. in Sepp and Rakhmonov (2023b).

□

Corollary 6.2 (*First-order solution for MGF*) *Dropping the residual function* $R^{[1]}$ *in Eq.* (107), *we obtain the first-order solution to the MGF of variable z as follows*

$$
G(\tau, T, z, v; \phi) = \exp\{-z\phi\} E^{[1]}(\tau, T, v; \phi). \tag{114}
$$

Proposition 6.1 *The MGF in* (114) *satisfies the condition*

$$
G(\tau, T, z, v; \phi)|_{\phi=0} = e^{-z\phi}
$$

and reproduces the expected value of z_T *exactly when* $\kappa_2 = 0$.

Proof By analogy to Propositions 4.3 and 4.5 in Sepp and Rakhmonov (2023b). □

In Appendix 5, we also provide the solution for $G(\tau, T, z, v; \phi)$ using the second order expansion with leading term $E^{[2]}(\tau, T, z; \phi)$. The second order expansion reproduces the variance of z exactly when $\kappa_2 = 0$. For practical purposes, we have found that the first-order solution for the MGF works fine under various market conditions, so we recommend using the first-order solution. In Figs. 2 and 3 we demonstrate the application of the first-order solution in Eq. (114) for computing the PDFs of the swap rate and the futures rate, respectively.

Corollary 6.3 (First-order expansion for the MGF of the swap rate) *MGF in Eq.* (114) *for the state-independent swap rate* s_T *in Eq.* (39), *defined by*

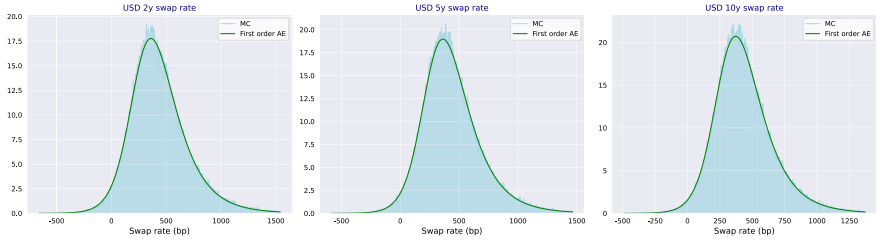


Fig. 2 PDFs computed using the inversion of the first-order solution $E^{[1]}$ in Eq. (108) for the state variable s_T at $T = 5$. The blue histogram is computed using realizations from MC simulations in model dynamics (9). We use the model parameters reported in Table 1

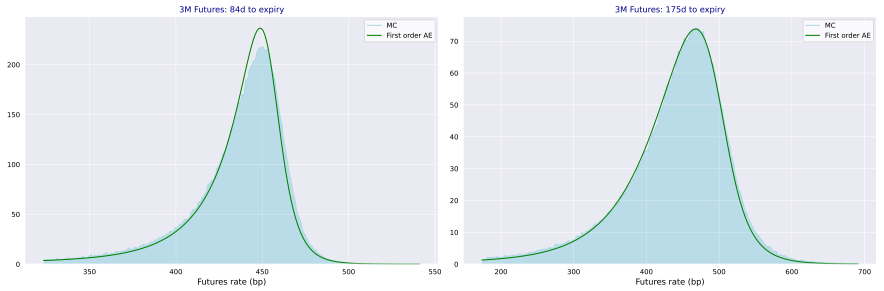


Fig. 3 PDFs computed using the inversion of the first-order solution $E^{[1]}$ in Eq. (108) for the three-month SOFR futures rate F_T. The blue histogram is computed using realizations from MC simulations in model dynamics (39). We use the model parameters reported in Table 3

$$G(\tau, T, s, v; \phi) = \mathbb{E}^A \left[e^{-\phi s_T} \middle| s_t = s, v_t = v \right]$$

can be decomposed as

$$G(\tau, T, s, v; \phi) = \exp\{-s\phi\} \left[E^{[1]}(\tau, T, v; \phi) + R^{[1]}(\tau, T, v; \phi) \right]. \tag{115}$$

The leading term $E^{[1]}$ and the remainder term $R^{[1]}$ can be identified by setting

$$\hat{a}_0(t) = a(t), \quad \hat{a}_1(t) \equiv 0, \quad \hat{b}(t) \equiv 0,$$
$$\hat{\kappa}_0(t) := \beta_2(t)\theta^2, \quad \hat{\kappa}_1(t) := \kappa_1 - \kappa_2\theta + 2(\kappa_2 - \beta_2(t))\theta, \quad \hat{\kappa}_2(t) := \kappa_2 - \beta_2(t)$$

everywhere in Theorem 6.1, where $a(t)$ and $\beta_2(t)$ are defined in Eq. (40).

Corollary 6.4 (*First-order solution for MGF of the swap rate*) *Dropping the residual function $R^{[1]}$ in Eq. (115), we obtain the first-order solution to the MGF of variable s as follows*:

$$G(\tau, T, s, v; \phi) = \exp\{-s\phi\} E^{[1]}(\tau, T, v; \phi). \tag{116}$$

In Fig. 2, we show the probability density function (PDF) computed using the inversion of the first-order solution $E^{[1]}$ in Eq. (116) for the $2y$, $5y$, $10y$ swap rate s_T observed at $T = 5$ in the dynamics (39). For comparison, the blue histogram shows realizations using MC simulations of the Nelson–Siegel model with log-normal SV (12) without the drift-freezing assumption in Eq. (37). We see that the first-order expansion consistently fits the histograms produced by the MC simulations.

Corollary 6.5 (Affine expansion for log-shifted futures price) *Let ζ_t denote log-shifted futures price defined by*

$$\zeta_t = \ln\left(F_t + \Delta^{-1}\right). \tag{117}$$

The moment generating function for ζ_T in (114)

$$G(\tau, T, \zeta, v; \phi) = \mathbb{E}^{\mathbb{Q}^T}\left[e^{-\phi\zeta_T}\middle|\zeta_t = \zeta, v_t = v\right]$$

can be decomposed as

$$G(\tau, T, \zeta, v, \phi) = e^{-\zeta\phi}\left[E^{[1]}(\tau, T, v; \phi) + R^{[1]}(\tau, T, v; \phi)\right]. \tag{118}$$

The leading term $E^{[1]}$ and the remainder term $R^{[1]}$ can be identified by setting

$$\begin{aligned}
&\hat{a}_0(t) = a_0(t), \quad \hat{a}_1(t) = a_1(t),\\
&\hat{b}(t) = a_0(t)^\top \eta(t) + \frac{1}{2}\left(\|a_0(t)\|^2 + a_1(t)^2\right),\\
&\hat{\kappa}_0(t) := -\vec{\beta}\eta\theta^2, \quad \hat{\kappa}_1(t) := \kappa_1 - \kappa_2\theta + 2(\kappa_2 + \vec{\beta}\eta)\theta, \quad \hat{\kappa}_2(t) := \kappa_2 + \vec{\beta}\eta,
\end{aligned}$$

everywhere in Theorem 6.1, where $a_0(t)$, $a_1(t)$, $\eta(t)$ are defined in (72) and (74).

Proof Applying Itô's lemma to Eq. (73), we can find the dynamics of ζ_t, v_t:

$$\begin{aligned}
d\zeta_t &= \left[-a_0(t)^\top \eta(t) - \frac{1}{2}\left(\|a_0(t)\|^2 + a_1(t)^2\right)\right](v_t + \theta)^2\, dt + (v_t + \theta)\left[a_0(t)^\top\, dW_t^{T,(0)} + a_1(t)\, dw_t^{T,(1)}\right],\\
dv_t &= (\hat{\kappa}_0(t) - \hat{\kappa}_1(t)v_t - \hat{\kappa}_2(t)v_t^2)dt + (v_t + \theta)\vec{\beta}(t)^\top\, dW_t^{T,(0)} + \beta_0(t)(v_t + \theta)dw_t^{T,(1)},
\end{aligned} \tag{119}$$

where $a_0(t)$, $a_1(t)$, $\eta(t)$ are defined in (72) and (74). Comparing the dynamics of ζ_t under T-forward measure in Eq. (119) to the dynamics in Eq. (102), we complete the proof. □

Corollary 6.6 (*First-order solution for MGF of the futures rate*) *Dropping the residual function $R^{[1]}$ in Eq. (118), we obtain the first-order solution to the MGF of variable ζ as follows*

$$G(\tau, T, \zeta, v; \phi) = \exp\{-\zeta\phi\}E^{[1]}(\tau, T, v; \phi). \tag{120}$$

In Fig. 3, we show the probability density function (PDF) computed using the inversion of the first-order solution $E^{[1]}$ in Eq. (120) for the futures rate F_T, settling

in $84d$ (Subplot A) and $175d$ (Subplot B) in the dynamics (73). For comparison, the blue histogram shows realizations based on MC simulations of factors X, Y in Nelson–Siegel model with log-normal SV (12). We see that the first-order expansion consistently fits the histograms produced by the MC simulations.

7 Implementation details

7.1 Valuation formulas

Our framework allows for generic implementation of any FHJM–SV specification using model specific MGF (assuming it is analytic). In particular:

- Swaptions are valued using formula (84) along with (80);
- Options on rate futures are valued using formula (92) along with (88).

When we use an FHJM–SV with a generic basis and log-normal SV driver in (12):

- Swaptions are valued using formula (84), assuming MGF in Eq. (80) is computed using first-order term $E^{[1]}$ of Eq. (116);
- Options on rate futures are valued using valuation formula (92), assuming MGF in Eq. (88) is computed using first-order expansion term $E^{[1]}$ in Eq. (120).

Under CIR SV model, we value swaptions and options on rate futures using Eqs. (100) and (101), respectively.

Our valuation formulas are valid for any choice of factor basis $B(\tau)$. In particular, we consider the implementation of FHJM–SV model (12) with Nelson–Siegel basis $B(\tau)$ specified in Eq. (21). Both implementations of log-normal SV and CIR SV model require solving the sytem of ODEs numerically because both dynamics have time-dependent parameters. On the ground of model implementation using MC simulation, we believe that log-normal SV model is the most robust for applications. Thus, as an example of implementation, we only consider FHJM–SV model with log-normal SV driver.

7.2 Numerical integration

Numerical evaluation of Fourier integrals in Eqs. (80) and (88) requires methods that work well for slowly decaying integrands and short expiries, that are typical for valuation of options on rate futures, though similar challenges may arise for valuation of swaptions as well. To improve the convergence of those integrals, we employ double exponential quadrature rule which is remarkably accurate for such kind of problems, see Andersen and Lake (2019) and Michalski and Mosig (2016).

Let us consider an integral

$$I = \int_0^\infty g(p)dp, \tag{121}$$

where

$$g(p) = \frac{1}{\pi}A(t)\Re\left[\frac{e^{\left(-\frac{1}{2}+ip\right)K}}{\left(-\frac{1}{2}+ip\right)^2}G(\tau,T,s,v;\phi=-\frac{1}{2}+ip)\right]$$

as in Eq. (80) for the swap rate or

$$g(p) = P(t,T)\frac{K+\Delta^{-1}}{\pi}\Re\left[e^{(-\frac{1}{2}+ip)k^*}\frac{1}{p^2+1/4}G(\tau,T,s,v;\phi=-\frac{1}{2}+ip)\right]$$

as in Eq. (88) for the futures rate. We apply exp–sinh transformation $p = k(t) = \exp\left(\frac{\pi}{2}\sinh t\right)$ and use trapezoidal rule for Eq. (121) to evaluate resulting integral as follows

$$I = \int_{-\infty}^{+\infty} g(k(t))k'(t)dt \approx h\sum_{n=-\infty}^{\infty} g(nh)k'(nh) \tag{122}$$

where h is the step size. Assuming we truncate the sum (122) over $n \in [-N_1, N_2]$, $N_1 > 0$, $N_2 > 0$, then DE rule looks as follows

$$\begin{aligned} I &\approx h\sum_{n=-N_1}^{n=N_2} w_n g(p_n), \\ p_n &= k(nh) = \exp\left(\frac{\pi}{2}\sinh(nh)\right), \\ w_n &= k'(nh) = \frac{\pi}{2}\cosh(nh)\exp\left(\frac{\pi}{2}\sinh(nh)\right), \end{aligned} \tag{123}$$

where p_n and w_n are abscissas and weights of the quadrature rule, respectively.

Given a grid of maturities $\{T_m\}$, $m = 1, \ldots, M$, we pre-calculate MGF G on a specified grid of ϕ up to T_M. Given computed G on the grid of ϕ at expiry T_m, we apply Eq. (121) to compute option values for all options observed at time T_m in succession.

7.3 Monte–Carlo simulation

For benchmarking option prices computed using numerical integration of analytic Fourier transform, we apply Monte Carlo (MC) simulation of FHJM–SV model with log-normal SV driver in Eq. (12). Applying Itô's formula to the logarithm of SV driver $\xi_t := \ln\sigma_t$, we obtain the following dynamics:

$$d\xi_t = \mu(\xi_t)\,dt + \vec{\beta}(t)^\top\, dW_t^{\mathbb{Q},(0)} + \beta_0(t) dw_t^{\mathbb{Q},(1)}, \quad \xi_0 = \ln\sigma_0,$$
$$\mu(\xi) := \left(-\kappa_1 + \kappa_2\theta - \frac{1}{2}\vartheta(t)^2\right) + \kappa_1\theta e^{-\xi} - \kappa_2 e^{\xi},$$

where $\vartheta(t)$ is defined in Eq. (68).

We consider time grid $\{t_k\}$, $t_{k+1} - t_k = \Delta_k$, $k \geq 0$. For MC simulation of log-driver ξ_t, we app backward Euler-Maruyama scheme defined as follows

$$\xi_{t_{k+1}} = \xi_{t_k} + \mu\left(\xi_{t_{k+1}}\right)(t_{k+1} - t_k) + \boldsymbol{\beta}(t_k)^\top\left(W_{t_{k+1}}^{\mathbb{Q},(0)} - W_{t_k}^{\mathbb{Q},(0)}\right) + \beta_0(t_k)\left(w_{t_{k+1}}^{\mathbb{Q},(1)} - w_{t_k}^{\mathbb{Q},(1)}\right). \tag{124}$$

where we assume that the time-dependent coefficients of model dynamics are constant over $t \in [t_k, t_{k+1}]$. Sepp and Rakhmonov (2023b) show that the Backward Euler–Maruyama scheme in (124) has a strong convergence rate of 1. Importantly, the scheme (124) for the log-driver ξ_t does not require any special treatment of the boundary points since ξ_t is defined on $(-\infty, +\infty)$. In contrast, MC simulation of CIR driver in Eq. (11) requires treatment of the boundary point at zero.

We summarize the discretization of the model factors in FHJM model (9) with log-normal SV (12) under risk-neutral measure as follows

$$\begin{aligned}
X_{t_{k+1}} &= e^{\mathbf{D}\Delta_k} X_{t_k} + \sigma_{t_k} e^{\mathbf{D}\Delta_k}\mathbf{C}(t_k)\left(W_{t_{k+1}}^{\mathbb{Q},(0)} - W_{t_k}^{\mathbb{Q},(0)}\right),\\
Y_{t_{k+1}} &= e^{\tilde{\mathbf{D}}\Delta_k} Y_{t_k} + \left(\sigma_{t_k}\right)^2 e^{\tilde{\mathbf{D}}\Delta_k}\tilde{\Omega}_{t_k}\Delta_k,\\
\xi_{t_{k+1}} &= \xi_{t_k} + \mu\left(\xi_{t_{k+1}}\right)(t_{k+1} - t_k) + \vec{\beta}(t_k)^\top\left(W_{t_{k+1}}^{\mathbb{Q},(0)} - W_{t_k}^{\mathbb{Q},(0)}\right) + \beta_0(t_k)\left(w_{t_{k+1}}^{\mathbb{Q},(1)} - w_{t_k}^{\mathbb{Q},(1)}\right),\\
\sigma_{t_{k+1}} &= e^{\xi_{t_{k+1}}}
\end{aligned}$$

where $X_{t_0} = X_0$, $Y_{t_0} = 0$, $\xi_{t_0} = \ln\sigma_0$, $W^{(0)}$ and $w^{(1)}$ are independent d-dimensional and one-dimensional Brownian motions, respectively. We note that for uniform time steps $\Delta_k \equiv \Delta$, matrix exponentials $e^{\mathbf{D}\Delta}$ and $e^{\tilde{\mathbf{D}}\Delta}$ need to be computed only once.

7.4 Estimation of Nelson–Siegel basis

For estimation of factors in Nelson–Siegel model in Sect. 2.3, we use U.S. Treasury yields data for fixed tenors of 3, 6, 12, 24, 36, 60, 84 and 120 months and estimate factors (x_1, x_2, x_3) using ordinary least squares regression following Diebold and Li (2006) from January 2007 to February 2024, using mean reversion $\lambda = 0.55$ as explained below. In Fig. 4 we plot the time series of 'fitted' as well 'empirical factors' in Nelson–Siegel model where we define the 'empirical' factors $(\hat{x}_1, \hat{x}_2, \hat{x}_3)$ as the $10y$ yield, the difference between the $10y$ and $3m$ yields and twice the $2y$ yield minus the sum of the $3m$ and $10y$ yields, respectively. Recall that in Nelson–Siegel model in Sect. 2.3 the formula for the bond yield $y_t(\tau)$, see Eq. (26), is[4]

[4] To preserve consistent interpretation of factors as in Diebold and Li (2006), we do not absorb the term λ^{-1} in x_3.

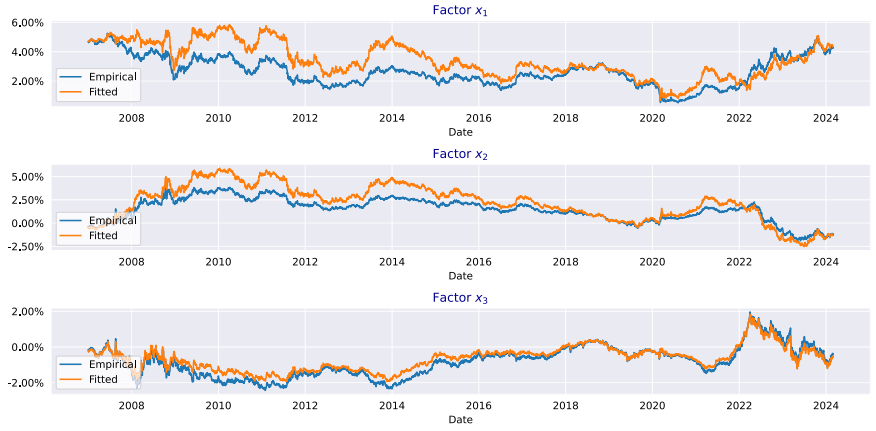


Fig. 4 Time series of 'estimated' and 'fitted' factors from January 2007 to February 2024, estimated by fitting the USD yield curve to Nelson–Siegel model. We use $\lambda = 0.55$

$$y_t(\tau) = x_1 + \frac{1 - e^{-\lambda\tau}}{\lambda\tau} x_2 + \frac{1 - e^{-\lambda\tau}(1 + \lambda\tau)}{\lambda\tau} \lambda^{-1} x_3$$

Given that:

$$\begin{aligned} y(10) &\approx x_1 + 0.18 x_2 + 0.17 \lambda^{-1} x_3 \\ y_t(10) - y_t(0.25) &\approx -0.75 x_2 + 0.11 \lambda^{-1} x_3 \\ 2y_t(2) - y_t(10) - y_t(0.25) &\approx 0.01 x_2 + 0.3 \lambda^{-1} x_3 \end{aligned}$$

we define 'fitted' factors $(\check{x}_1, \check{x}_2, \check{x}_3)$ through factors of the Nelson–Siegel model by setting $\check{x}_1 = x_1$, $\check{x}_2 = -x_2$, $\check{x}_3 = 0.3\lambda^{-1} x_3$. We show in Fig. 4, 'fitted' factors $(\check{x}_1, \check{x}_2, \check{x}_3)$ can be naturally interpreted as 'level', 'slope' and 'curvature'.

7.5 Model calibration to swaptions

We fix $\{2y, 5y, 10y\}$ as benchmark tenors. For a given day we use estimates $\hat{X}_t$ obtained as in previous section. As we noted in Sect. 9, it is natural to parametrize volatility matrix $\mathbf{C}(t)$ through correlation of benchmark yields $\mathbf{R}$ and their volatilities $\mathbf{A}(t)$. We estimate the correlation between benchmark yields historically and set it to

$$\mathbf{R} = \begin{pmatrix} 1 & 0.99 & 0.98 \\ 0.99 & 1 & 0.97 \\ 0.98 & 0.97 & 1 \end{pmatrix}.$$

We also set mean reversion to $\lambda = 0.55$. This value is within empirically observed range $\lambda \in [0.5, 1]$ for the USD market data, see Christensen et al. (2009). We have observed that using higher values of λ requires excessively time-inhomogeneous correlation matrix $\mathbf{R}$ to fit observed swaption implied volatilities across expiries.

To keep our presentation as parsimonious as possible, we thus choose $\lambda = 0.55$. We can calibrate the correlation matrix $\mathbf{R}$ to market data of spread options, if this is available.

Once correlation matrix $\mathbf{R}$ is fixed, we reduce the calibration of term-structure volatility matrix $\mathbf{C}(\cdot)$ to fitting volatilities $\vec{a}(\cdot) = (a_1(\cdot), a_2(\cdot), a_3(\cdot))^\top$ of benchmark yields, following parametrization in Eq. (130). Thus, we fit term-structure of parameters $\vec{a}(\cdot)$, $\vec{\beta}(\cdot)$, $\beta_0(\cdot)$ to market prices of swaptions. After fixing the expiries of traded swaptions to $T_1^* < T_2^* < \cdots < T_n^*$, we perform the calibration by bootstrapping, assuming that model parameters $\vec{a}(t)$, $\vec{\beta}(t)$, $\beta_0(t)$ are flat over $t \in (T_{k-1}^*, T_k^*]$, $k = 1, \ldots, n$, $T_0^* = 0$.

We fit the set of parameters at time T_k^* to available instruments expiring at T_k^*. For each expiry T, we minimize the squared difference between model and market implied volatilities as follows

$$WMSE = \sum_n w_n(T,K)\big(\sigma_n^{model}(T,K) - \sigma_n^{implied}(T,K)\big)^2, \tag{125}$$

weighted by corresponding vega, $w_n(T,K) = \mathcal{V}_n(T,K)$, which are computed using Bachelier model. During the calibration we enforce the restriction in Eq. (33) on κ_2.

In Table 1 we show calibrated model parameters for USD swaptions using market data as of mid-August 2023. We use four swaption slices $T_k^* \in \{1y, 2y, 3y, 5y\}$ per tenor, where each swaption slice consist of five options with equally spaced strikes, centered on at-the-money forward swap rate. We set $\kappa_1 = 0.25$, $\kappa_2 = 0.5$, the choice that ensures that volatility of volatility remains relatively stable across expiries. We note that achieving such stability becomes non-trivial in multi-factor case as volatility beta inherently depends on choice of factor correlation $\mathbf{R}$ (Fig. 5).

In Fig. 6, we compare the model implied normal volatilities for swaption with 10y expiry, computed using the first-order affine expansion of Eq. (116). As a benchmark, we use Nelson–Siegel term structure model where model factors (9) are simulated under risk-neutral measure with 200,000 trials and daily time steps in MC simulation. We show the 95% confidence interval of MC estimates as "bid-ask" bars and use the model parameters for options in USD market reported in Table 1. As seen, option values computed using the first-order affine expansion are consistent with MC simulations.

Table 1 Parameters of Nelson–Siegel term-structure model with log-normal SV driver (12) calibrated to USD swaption market data. Expiry is shown in the first column and underlying swap tenors are $\{2y, 5y, 10y\}$

Expiry, T^*	$a_1(t)$	$a_2(t)$	$a_3(t)$	$\beta_1(t)$	$\beta_2(t)$	$\beta_3(t)$	$\beta_0(t)$
1y	0.0143	0.0129	0.0113	0.0705	− 0.3236	0.7477	0.0300
2y	0.0128	0.0129	0.0112	0.5837	0.5938	− 0.5586	0.1067
3y	0.0113	0.0132	0.0113	− 0.0093	0.3647	− 0.2500	0.0757
5y	0.0055	0.0097	0.0086	0.7293	− 0.4866	− 0.1628	0.0715

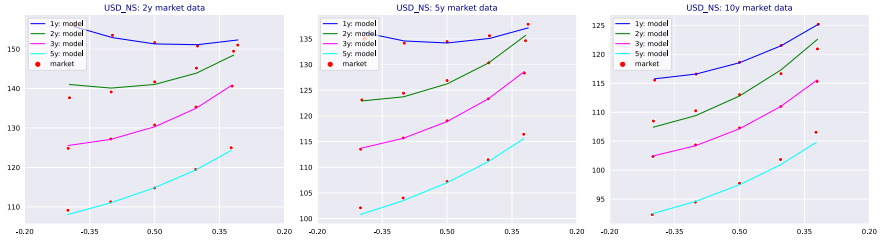


Fig. 5 Accuracy of model calibration to USD swaptions for $\{2y, 5y, 10y\}$ tenors, reported in Table 1 using four slices per tenor. Solid lines represent model implied normal volatilities computed using first-order affine expansion in Eq. (108). Dots display market implied volatilities

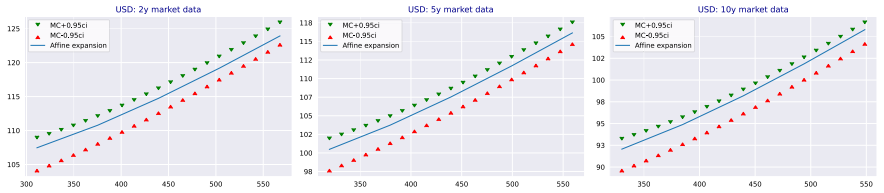


Fig. 6 Implied normal volatilities of USD swaptions for $\{2y, 5y, 10y\}$ tenors with $5y$ expiry and uniform range of strikes computed using first-order affine expansion (108) and MC simulations using calibrated parameters to USD swaptions data reported in Table 1. Dashed lines labeled by $MC - 0.95ci$ and $MC + 0.95ci$ are MC 95% confidence intervals computed using MC simulations

Table 2 Description of shocks of model parameters applied for producing Fig. 7

Scenario	Shock $\vec{\beta}$	Shock β_0	Shock $\vec{a}$
1	Base value	Base value	Base value
2	Base value	Base value	Shift $\vec{a}$ up by 0.02
3	Base value	Multiply β_0 by 4	Base value
4	Multiply $\vec{\beta}$ by -2	Base value	Base value

7.6 Accuracy of affine expansion under shocks

In Fig. 7, we show the quality of affine expansion (108) using a set of shocked model parameters. Our objective is to demonstrate that the affine expansion remains accurate under various shocks. As a base scenario, we use flat parameters set as follows:

$$\vec{\beta}(t) = (0.2, 0.2, 0.2)^{\top},\ \beta_0 = 0.2,\ \vec{a} = (0.01, 0.01, 0.01).$$

In Table 2, we show model parameters for scenario 1 corresponding to the initial parameter set; scenario 2 corresponding to increasing the volatility level by 0.02 (interpreted as increasing ATM volatilities by 2%); scenario 3 with volatility of volatility parameter β_0 increase fourfold (increasing the convexity of implied volatilities);

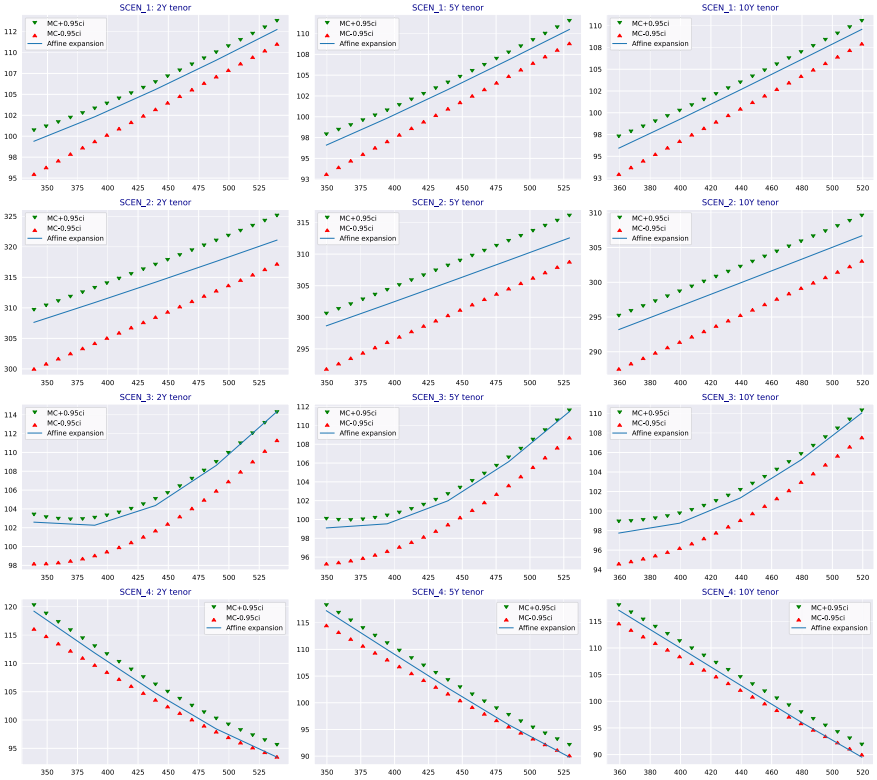


Fig. 7 Swaption approximation (108), benchmarked against MC simulations. Dashed lines labeled by $MC - 0.95ci$ and $MC + 0.95ci$ are MC 95% confidence intervals computed using MC simulations. Solid blue line represent implied normal volatilities calculated using first-order affine expansion. We fix swaption expiry to $5y$ and swaption tenors are set to $2y$, $5y$, $10y$ in first, second and third columns, respectively. The four scenarios are described in Table 2

scenario 4 with $\vec{\beta}(t)$ multiplied by -2 (inverting and increasing the skeweness of implied volatilities).

We see that the affine expansion may be less accurate, compared to Monte-Carlo mean estimates, in out-of-the-money region when the magnitude of rate-volatility betas is large or volatility-of-volatility is very high. However, for these scenarios, it is important to take into account additional noise introduced by Monte-Carlo errors. Overall, the affine approximation for swaption valuation using the first-order affine expansion demonstrates excellent accuracy in most cases. We mention that the first-order affine expansion is usually sufficient for practical purposes. To improve the accuracy, we can employ the second-order expansion, see Appendix 5.

Table 3 Parameters of Nelson–Siegel term-structure model with log-normal SV driver (12) calibrated to 3M SOFR options market data. Expiry is shown in the first column

Expiry, T^*	$a_1(t)$	$a_2(t)$	$a_3(t)$	$\beta_1(t)$	$\beta_2(t)$	$\beta_3(t)$	$\beta_0(t)$
75d	0.0049	0.0024	0.0005	− 1.995	0.99	0.0	1.449
103d	0.0041	0.0017	0.0039	− 1.005	0.502	0.0	0.213

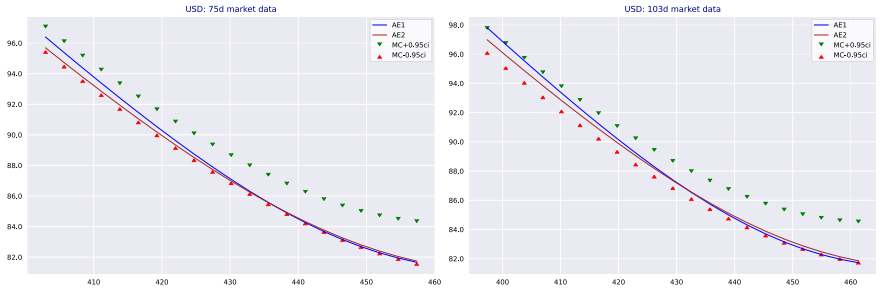


Fig. 8 Accuracy of model calibration to 3M SOFR options, reported in Table 3 using two slices, having 75 and 103 days to expiry. Solid lines represent model implied normal volatilities computed using first-order affine expansion in Eq. (108). Orange dots display market implied volatilities

7.7 Model calibration to SOFR options

Now we demonstrate the quality of calibration to 3M SOFR options, which are most liquidly traded short-term interest rate options in the USD market. Similar to swaptions, we use Nelson–Siegel term structure model. We further use scalar structure for the volatility beta $\vec{\beta} = \left(\beta, -\frac{\beta}{2}, 0\right)^\top$ to avoid over-parametrization. This structure of $\vec{\beta}$ is based on historical analysis of covariance matrix between Nelson–Siegel factors and short-term rates volatilities. Similarly, we impose the restriction on κ_2 during the calibration.

For illustration, we use market data as of October 2023 and calibrate the model to the two most liquid 3M SOFR options with expiries $T_k^* \in \{84\text{d}, 175\text{d}\}$. For each slice, we take five strikes, centered on settlement value of underlying futures for the previous day. In Table 3 we show calibrated model parameters and in Fig. 8 we show the quality of calibration. Here, option strikes are reported in normal delta space, where we include puts and calls with deltas in the range $[-0.25, -0.5)$ and $[0.5, 0.25]$, respectively. In addition to market implied volatilities, we display volatilities for one tick change in option price as a proxy for "bid-ask" bars.

We see that implied volatilities of SOFR options exhibit extremely high convexity, as well as skew. For that reason, we set $\kappa_1 = 0.5$, $\kappa_2 = 1.0$ to account for speed of

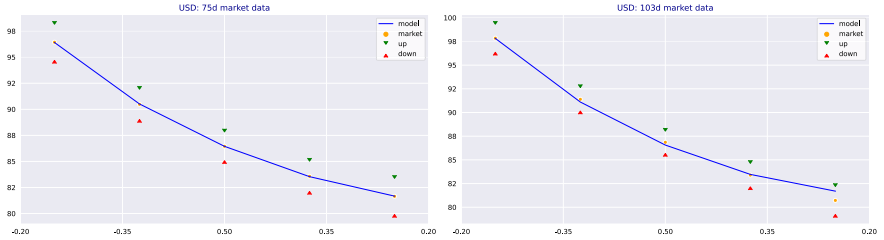


Fig. 9 Implied normal volatilities of SOFR options with 75d and 103d to expiry and uniform range of strikes computed using first-order affine expansion (108) and MC simulations using parameters calibrated to SOFR options, see Table 3. Dashed lines labeled by $MC - 0.95ci$ and $MC + 0.95ci$ are MC 95% confidence intervals computed using MC simulations

convexity decay and make sure that volatility of volatility remains relatively stable across different slices.[5]

In Fig. 9, we compare the model implied normal volatilities for 3M SOFR options with 75 and 103 days to expiry, computed using the first-order affine expansion of Eq. (120). As a benchmark, we use Nelson–Siegel term structure model where model factors (9) are simulated under risk-neutral measure with 200,000 trials and daily time steps in MC simulation. We show the 95% confidence interval of MC estimates as "bid-ask" bars and use model parameters for the USD market as reported in Table 3.

8 Conclusions

We have extended the Factor Heath–Jarrow–Morton (FHJM) framework to incorporate a stochastic volatility (SV) driver correlated with curve factors. For practical purposes, we implemented two SV drivers: CIR driver and the log-normal driver. Regarding the choice of factor loadings, we have applied Nelson–Siegel term structure model, see Nelson and Siegel (1987), widely used by central banks for modeling the term structure under physical measure $\mathbb{P}$. Following Lyashenko and Goncharov (2023), we augmented the main stochastic factor by locally deterministic factor to ensure the model is arbitrage-free under pricing measure $\mathbb{Q}$. In addition to Nelson–Siegel basis, more general exponential basis has been considered for factor loading, which allows for flexible modeling of the yield curve in multi-factor setting.

We derived the dynamics of swap rate and interest rate futures in general FHJM–SV framework and further derived convexity adjustment under Gaussian FHJM and FHJM–SV frameworks with CIR and the log-normal SV drivers.

[5] Furthermore, high values of κ_1, κ_2 are necessary to satisfy the regularity conditions on moments of the volatility process. In our case, we check that low-dimensional truncation of infinite-dimensional matrix $\Lambda^{(0,\infty)}$ for the solution of moments of the volatility process is stable, i.e. its eigenvalues have negative real parts, see Sepp and Rakhmonov (2023b) for further details.

We also derived analytical solution for the moment generating function (MGF) in FHJM–SV model with CIR driver and accurate approximate solution for the MGF in FHJM–SV model with the log-normal driver. Our analytical approach allows valuation of options on swap rate (swaptions) and options on futures rate, which is analytic up to the inversion of Fourier transform.

As a practical example, assuming Nelson–Siegel basis in FHJM–SV model with log-normal SV driver, its calibration to market data of swaptions on three benchmark tenors (2y, 5y and 10y) and to SOFR options demonstrates that the model can accurately fit each market using term structure of three model parameters: level, skew, and convexity for controlling model implied volatilities. Proposed model yields accurate values for swaptions and options on SOFR futures when compared to Monte-Carlo simulations. We also established that affine expansion remains accurate under various shocks of the model parameters.

Given popularity of Nelson–Siegel term structure model, proposed model offers valuable toolkit for building joint scenarios of the shape of both the yield curve and the implied volatilities, as well as for risk-management of fixed-income derivatives. As a future extension, we shall consider a multi-factor SV driver for FHJM–SV model which can incorporate the dynamics of the interest rate skew and convexity.

Appendix 1: Volatility parametrization

Let us discuss parametrization of the volatility matrix $\mathbf{\Sigma}_t$ in (4) for arbitrary basis $B(\tau)$. We consider the approach outlined in Lyashenko and Goncharov (2023) and Andersen and Piterbarg (2010). We set the number of factors to d, choose d key term points $\tau_1 < \cdots < \tau_d$ and introduce d-dimensional vector of continuously compounded benchmark yields

$$y_t = \big(y_t(\tau_1), \dots, y_t(\tau_d)\big)^\top,$$

where

$$y_t(\tau_k) = \frac{1}{\tau_k} \int_0^{\tau_k} f_t(\tau)\, d\tau, \quad k = 1, \dots, d.$$

We begin with the most generic setting, when individual yields are assumed to have following dynamics

$$dy_t(\tau_k) = (\dots)\, dt + \sigma_t^{(k)} \big(a_k(t) + b_k(t) y_t(\tau_k)\big)\, dZ_t^{(k)}, \quad k = 1, \dots, d, \tag{126}$$

where $a_k(t)$ defines instantaneous volatility, $\sigma_t^{(k)}$ is SV driver for $y_t(\tau_k)$ and $Z_t = \left(Z_t^{(1)}, \dots, Z_t^{(d)}\right)^\top$ is a d-dimensional Brownian motion with correlation matrix $\mathbf{R}$. The matrix $\mathbf{R}$ is an input and can be estimated, e.g. using econometric analysis.

As seen, individual yields have displaced diffusion SV dynamics in (126). Using vector notation, we can conveniently rewrite (126) as

$$dy_t = (\dots)\,dt + \hat{\mathbf{\Sigma}}_t \mathbf{A}(t)\,dZ_t, \tag{127}$$

where for future reference we introduced

$$\hat{\mathbf{\Sigma}}_t := \operatorname{diag}\left(\sigma_t^{(1)}, \dots, \sigma_t^{(d)}\right),$$
$$\mathbf{A}(t) := \operatorname{diag}\left(a_1(t) + b_1(t) y_t(\tau_1), \dots, a_d(t) + b_d(t) y_t(\tau_d)\right)$$

Using the dynamics of benchmark yields y_t in (127), we assume following parametrization for matrix $\mathbf{\Sigma}_t$:

$$\mathbf{\Sigma}_t = \hat{\mathbf{\Sigma}}_t \mathbf{C}(t; X_t, Y_t).$$

Thus, when factors dynamics are given by (4), the dynamics of $y_t(\tau_k)$ in (4) become

$$dy_t(\tau_k) = (\dots)\,dt + \sigma_t^{(k)} \left[\frac{1}{\tau_k}\int_0^{\tau_k} B(\tau)\,d\tau\right] \mathbf{C}(t; X_t, Y_t)\,dW_t.$$

The above equations can be rewritten as

$$dy_t = (\dots)\,dt + \hat{\mathbf{\Sigma}}_t \mathbf{B_Y} \mathbf{C}(t; X_t, Y_t)\,dW_t. \tag{128}$$

where matrix $\mathbf{B_Y}$ is formed by stacking vectors $\frac{1}{\tau_k}\int_0^{\tau_k} B(\tau)\,d\tau$ for key term points τ_k row-by-row. By comparing Eqs. (127) and (128), we see that volatility matrix $\mathbf{C}(t; X_t, Y_t)$ should be chosen as

$$\mathbf{C}(t; X_t, Y_t) = \mathbf{B_Y}^{-1} \mathbf{A}(t) \mathbf{R_C}, \tag{129}$$

where $\mathbf{R_C}$ is a lower-triangular $d \times d$ matrix found by Cholesky decomposition of the correlation matrix $\mathbf{R}$. We note that $\mathbf{C}(t; X_t, Y_t)$ is linear in $\{X_t, Y_t\}$ as

$$a_k(t) + b_k(t) y_t(\tau_k) = a_k(t) - b_k(t)\tau_k^{-1} \ln P\left(t, t+\tau_k; X_t, Y_t\right)$$
$$= a_k - b_k \tau_k^{-1} \ln\left(\frac{P(0, t+\tau_k)}{P(0,t)}\right) + b_k \tau_k^{-1} B_P(\tau_k) X_t + b_k \tau_k^{-1} \tilde{B}_P(\tau_k) Y_t.$$

Consequently, auxiliary vector Ω_t becomes quadratic in X_t and Y_t, see (5), which implies that the drift of vector dY_t must contain terms quadratic in Y_t. Regardless of the dynamics of d-factor volatility $\mathbf{\Sigma}_t$ and its correlation to curve factor X_t, such specification is prone to explosions, therefore we argue that modeling interest skews in the FHJM model is only possible when $b_k(t) \equiv 0$, i.e. curve driver X_t, conditional on the SV driver, is normal. Such assumption makes matrix $\mathbf{C}(t)$ deterministic, henceforth we set

$$\mathbf{C}(t) = \mathbf{B_Y}^{-1} \mathbf{A}(t) \mathbf{R_C}, \tag{130}$$

where $\mathbf{A}(t) := \operatorname{diag}\big(a_1(t), \dots, a_d(t)\big)$. Parametrization (130) allows to conveniently express volatility matrix in terms of correlation between benchmark yields and their instantaneous volatilities, which is important for the purposes of model calibration.

Appendix 2: Pure exponential basis

We note that, in general, the dimension of the auxiliary factor Y_t grows quadratically with d, where d is the dimension of the main factor X_t. As an alternative which results in a linear growth of dimension of Y_t, we consider *pure exponential* (PE) basis with spectral set $\Lambda = \{\lambda_1, \dots, \lambda_d\}$, specified as follows:

$$B(\tau) = \big(e^{-\lambda_1 \tau}, \dots, e^{-\lambda_d \tau}\big), \quad \lambda_k = \lambda_{\min} + \frac{k-1}{d-1}\big(\lambda_{\max} - \lambda_{\min}\big), \quad k = 1, \dots, d, \tag{131}$$

where $0 < \lambda_{\min} < \lambda_{\max}$. In this case, the generating matrix $\mathbf{D}$ of the main basis $B(\tau)$ is diagonal

$$\mathbf{D} = \operatorname{diag}(-\Lambda) = \begin{pmatrix} -\lambda_1 & & \\ & \ddots & \\ & & -\lambda_d \end{pmatrix} \tag{132}$$

and auxiliary basis $\tilde{B}(\tau)$ is set as pure exponential with spectral set $\tilde{\Lambda}$ and generating matrix $\tilde{\mathbf{D}}$, given by

$$\begin{aligned} &\tilde{\Lambda} = \Lambda \cup \{\Lambda + \Lambda\}, \quad \Lambda = \{\lambda_1, \dots, \lambda_d\}, \quad \Lambda + \Lambda = \{\lambda_i + \lambda_j \,|\, i,j = 1, \dots, d\}, \\ &\tilde{\mathbf{D}} = \operatorname{diag}\big(-\tilde{\Lambda}\big), \end{aligned} \tag{133}$$

see Lyashenko and Goncharov (2023). Thus, vector $\tilde{\Omega}_t$ in (10) can be found by rearranging the terms:

$$\begin{aligned} \tilde{B}(\tau)\tilde{\Omega}_t &= B(\tau)\mathbf{C}(t)\mathbf{C}(t)^\top \left(\int_0^\tau B(u)\,du\right)^\top \\ &= \big(e^{-\lambda_1 \tau}, \dots, e^{-\lambda_d \tau}\big)\mathbf{C}(t)\mathbf{C}(t)^\top \left(\frac{1-e^{-\lambda_1 \tau}}{\lambda_1}, \dots, \frac{1-e^{-\lambda_d \tau}}{\lambda_d}\right)^\top. \end{aligned} \tag{134}$$

As shown in Lyashenko and Goncharov (2023), the dimensionality of spectral set $\tilde{\Lambda}$ is usually $n = 3d - 1$. As a small refinement, we calculate the dimensionality of the auxiliary basis for any possible values of $\lambda_{\min}$, $\lambda_{\max}$. We omit its proof as it involves direct calculations.

Proposition 10.1 *We let* $\bar{\Delta} := \frac{1}{d-1}\big(\lambda_{\max} - \lambda_{\min}\big)$. *If* $\lambda_{\min} \notin \{\bar{\Delta}, 2\bar{\Delta}, \dots, (d-1)\bar{\Delta}\}$, *then the dimensionality of auxiliary spectral set is* $n = 3d - 1$. *If* $\lambda_{\min} = k\bar{\Delta}$ *for some* $k = 1, \dots, d-1$, *then* $n = k + 2d - 1$.

Thus, without loss of generality, we choose $\lambda_{\min}$ and $\lambda_{\max}$ such that $n = 3d - 1$. In this case, it is guaranteed that $\Lambda \cap \{\Lambda + \Lambda\} = \varnothing$. Therefore, to calculate the vector process in (134), we resolve $\tilde{\Omega}_t$ from the equation

$$\tilde{B}(\tau)\tilde{\Omega}_t = \left(e^{-\lambda_1\tau}, \dots, e^{-\lambda_d\tau}\right)\mathbf{C}(t)\mathbf{C}(t)^\top\left(\frac{1-e^{-\lambda_1\tau}}{\lambda_1}, \dots, \frac{1-e^{-\lambda_d\tau}}{\lambda_d}\right)^\top.$$

If we denote

$$\tilde{\Omega}_t := \left(\tilde{\Omega}_t^1, \tilde{\Omega}_t^2, \dots, \tilde{\Omega}_t^m\right)^\top, \quad \mathbf{M}(t) := \mathbf{C}(t)\mathbf{C}(t)^\top, \quad \mathbf{M}(t) = \{m_{kj}(t)\}_{k,j=1,\dots,d},$$

by rearranging the terms we can find that

$$\tilde{\Omega}_t^k = \begin{cases} \sum_{j=1}^d \frac{m_{kj}}{\lambda_j}, & \lambda_k \in \Lambda, \\ -\sum_{\substack{i,j=1: \\ i+j=k}}^d \frac{m_{ij}}{\lambda_j}, & \lambda_k \in \Lambda + \Lambda. \end{cases} \tag{135}$$

Corollary 10.1 *We consider an FHJM model with PE basis $B(\tau)$ given by (131). Then dynamics of model factors $\{X_t, Y_t\}$ is given by (9), where generating matrix of the main basis $B(\tau)$, auxiliary basis $\tilde{B}(\tau)$ and vector $\tilde{\Omega}_t$ are given by (132), (133) and (135), respectively. Instantaneous forward rates $f_t(\tau)$ are then given by*

$$f_t(\tau) = B(\tau)X_t + \tilde{B}(\tau)Y_t + \hat{f}_t(\tau), \quad \hat{f}_t(\tau) = f_0(t+\tau).$$

Prices of zero-coupon bonds are given by

$$P(t, t+\tau) = P(t, t+\tau; X_t, Y_t) = \frac{P(0, t+\tau)}{P(0,t)} \exp\left(-B_P(\tau)X_t - \tilde{B}_P(\tau)Y_t\right), \tag{136}$$

where

$$B_P(\tau) = \left(e^{-\lambda_1\tau}, \dots, e^{-\lambda_d\tau}\right), \quad \tilde{B}_P(\tau) = \left(\frac{1-e^{-\lambda_1\tau}}{\lambda_1}, \dots, \frac{1-e^{-\lambda_d\tau}}{\lambda_d}\right).$$

Proof Follows from Theorem 2.1. □

Appendix 3: Dynamics of swap and futures rates

Proof of Theorem 3.1

As shown below, the key part of the proof is to show the validity of the measure change from $\mathbb{Q}$ to $\mathbb{Q}^A$. Our argument is very close to the one used in Sepp and Rakhmonov (2025), which, in turn, utilizes idea from Carr et al. (2014) and Ruf (2015). To convey the important aspects of proof, we assume the log-normal dynamics (12)

for the SV. The proof carries over almost literally to the CIR SV and we omit it for brevity.

Proof If we perform a change of measure from risk-neutral $\mathbb{Q}$ to annuity measure $\mathbb{Q}^A$, the corresponding Radon–Nikodym derivative is given by [see Lucic (2016)]

$$Z_t := \left.\frac{d\mathbb{Q}^A}{d\mathbb{Q}}\right|_{\mathcal{F}_t} = \frac{A(t)/A(0)}{N(t)}, \tag{137}$$

where $N(t) = \exp\left(\int_0^t r_u du\right)$ is a money market account using short rate r_t. We find the $\mathbb{Q}$-dynamics of A as follows

$$dA(t) = \frac{\partial A(t)}{\partial t}dt + \left(\nabla_X A(t)\right)^\top dX_t + \left(\nabla_Y A(t)\right)^\top dY_t + \sigma_t^2 \frac{1}{2} Tr\left\{\mathbf{C}(t)^\top \nabla_X^2 A(t)\mathbf{C}(t)\right\}dt. \tag{138}$$

Then, the dynamics of annuity A in (138) can be rewritten as

$$dA(t) = A(t)\left[(\ldots)dt + \sigma_t \frac{1}{A(t)}\left(\nabla_X A(t)\right)^\top \mathbf{C}(t)\, dW_t^{\mathbb{Q},(0)}\right] = A(t)\left[(\ldots)dt + \sigma_t L_X(t)^\top \mathbf{C}(t)\, dW_t^{\mathbb{Q},(0)}\right].$$

Thus, the dynamics of Radon-Nikodym derivative in Eq. (137) is

$$Z_t := \mathcal{E}\left(\xi_t\right), \quad \xi_t := \int_0^t \sigma_u L_X(u)^\top \mathbf{C}(t)\, dW_u^{\mathbb{Q},(0)}$$

where $\mathcal{E}(\cdot)$ denotes Itô exponential. We consider sequence of stopping times $\tau_n \uparrow +\infty$, where $\tau_n := \inf\{t : \langle \xi, \xi\rangle_t \geq n\}$ and introduce measure $\mathbb{Q}^n$, indexed by $n \geq 1$, defined as

$$\left.\frac{d\mathbb{Q}^{(n)}}{d\mathbb{Q}}\right|_{\mathcal{F}_t} = \mathcal{E}\left(\xi_{\tau_n \wedge t}\right)$$

It is possible to show that Z_t is a proper martingale iff

$$\lim_{n\to\infty} \mathbb{Q}^{(n)}(\tau_n \leq t) = 0. \tag{139}$$

Since $\mathbb{Q}^{(n)}(\tau_n \leq t) = \mathbb{Q}^{(n)}(\langle \xi, \xi\rangle_t \geq n)$, the quadratic variation of ξ_t equals

$$\begin{aligned} \langle \xi, \xi\rangle_t &= \int_0^t \sigma_s^2 L_X(u)\mathbf{C}(u)\mathbf{C}(u)^\top L_X(u)^\top du, \\ d\sigma_t &= \left[\kappa_1\theta - (\kappa_1 - \kappa_2\theta)\sigma_t - \left(\kappa_2 - \vec{\beta}(t)^\top \mathbf{C}(t)^\top L_X(t)\right)\sigma_t^2\right] dt \\ &\quad + \sigma_t\left(\vec{\beta}(t)^\top dW_t^{n,(0)} + \beta_0(t) dw_t^{n,(1)}\right) \end{aligned} \tag{140}$$

where $W_t^{n,(0)}$, $w^{n,(1)}$ are Brownian motions under $\mathbb{Q}^{(n)}$, $t \le \tau_n$. Since L_X in (30) admits estimate[6]

$$\min_{n=0,\dots,N-1} B_P(T_{n+1} - s) \le L_X(u) \le \max_{n=0,\dots,N-1} B_P(T_{n+1} - s)$$

we can bound $\langle \xi, \xi \rangle_t$ in (140) from above as follows

$$\mathbb{E}^{(n)} \langle \xi, \xi \rangle_t \le C \times \mathbb{E}^{(n)} \left(\sup_{u \in [0,t]} \sigma_u^2 \right) \tag{141}$$

where constant C does not depend on n. Using the $\mathbb{Q}^{(n)}$-dynamics of σ_t, it follows from the Proposition 3.2 of Sepp and Rakhmonov (2023b), that the right-hand side of (141) is finite as soon as condition (33) holds. Now, using Markov inequality we verify (139) and prove that Z_t is a proper martingale, therefore $\mathbb{Q} \sim \mathbb{Q}^A$. By Girsanov theorem, the process

$$W_t^{A,(0)} := W_t^{(0)} - \int_0^t \sigma_u \mathbf{C}(t)^\top L_X(u)\, du$$

is a Brownian Motion under $\mathbb{Q}^A$. Substituting it into Eq. (12), we derive the dynamics of factors $\{X_t, Y_t\}$ as in Eq. (32). The $\mathbb{Q}^A$-dynamics in model (9) with log-normal driver (11) becomes

$$\begin{aligned} d\sigma_t &= \big(\kappa_1 + \kappa_2 \sigma_t\big)(\theta - \sigma_t)dt + \sigma_t \Big(\vec{\beta}(t)^\top \Big[dW_t^{A,(0)} + \sigma_t \mathbf{C}(t)^\top L_X(t) dt \Big] + \beta_0(t) dw_t^{A,(1)} \Big) \\ &= \Big[\kappa_1 \theta - (\kappa_1 - \kappa_2 \theta)\sigma_t - \Big(\kappa_2 - \vec{\beta}(t)^\top \mathbf{C}(t)^\top L_X(t) \Big) \sigma_t^2 \Big] dt + \\ &\quad + \sigma_t \vec{\beta}(t)^\top dW_t^{A,(0)} + \beta_0(t) \sigma_t\, dw_t^{A,(1)} \end{aligned}$$

□

Proof of Theorem 3.6

Proof If we change the measure from risk-neutral $\mathbb{Q}$ to T_E-forward measure $\mathbb{Q}^{T_E}$, the Radon–Nikodym derivative for the measure change equals

$$\left. \frac{d\mathbb{Q}^{T_E}}{d\mathbb{Q}} \right|_{\mathcal{F}_t} = \frac{P(t, T_E)/P(0, T_E)}{N(t)} = \mathcal{E}\Big(- \int_0^t \sigma_s \eta(s)^\top dW_s^{\mathbb{Q},(0)} \Big), \tag{142}$$

[6] Allowing slight abuse of notations, we use vector notation to compare vectors components-wise. For example, $X < Y$ is used to denote $X_i < Y_i$ for all $i = 1, \dots, N$, where $X, Y \in \mathbb{R}^N$.

where $\eta(\cdot)$ is defined in (74). Expression (142) for the Radon–Nikodym derivative can be easily deduced from the expression for the volatility of the instantaneous forward rate in model (9). We note that the proof that $\mathbb{Q} \sim \mathbb{Q}^{T_E}$, which is required for the measure change, is carried out using same arguments as in Appendix 3.1, with one difference that we use $\xi_t := -\int_0^t \sigma_u \eta(u)^\top \, dW_u^{\mathbb{Q},(0)}$

Therefore, by Girsanov theorem, the process $W_t^{T_E,(0)} := W_t^{\mathbb{Q},(0)} + \int_0^t \sigma_s \eta(s)\, ds$ is a Brownian motion under $\mathbb{Q}^{T_E}$. Substituting into Eq. (71) we derive the dynamics of futures rate in (73). Finally, the $\mathbb{Q}^{T_E}$-dynamics of the volatility process becomes as follows

$$\begin{aligned} d\sigma_t &= \big(\kappa_1 + \kappa_2 \sigma_t\big)(\theta - \sigma_t)dt + \sigma_t \vec{\beta}(t)^\top (dW_t^{T_E,(0)} - \sigma_t \eta(t) dt) + \beta_0(t)\sigma_t \, dw_t^{T_E,(1)} \\ &= [\kappa_1 \theta - (\kappa_1 - \kappa_2 \theta)\sigma_t - (\kappa_2 + \vec{\beta}(t)^\top \eta(t))\sigma_t^2]\, dt + \sigma_t \vec{\beta}(t)^\top \, dW_t^{T_E,(0)} + \beta_0(t)\sigma_t \, dw_t^{T_E,(1)} \end{aligned}$$

□

Appendix 4: Futures convexity adjustment

Proof to Theorem 3.3

For Guassian FHJM model it is easier to devise a direct proof that utilizes the deterministic nature of rates volatilities. We use following two relationships, that can be found in Henrard (2005) in context of one-factor HJM model. Its extension into multi-dimensional setting is straightforward and we skip its proof:

$$\begin{aligned} e^{\int_{T_S}^{T_E} r(u)\, du} &= P(T_S, T_E)^{-1} e^{\frac{1}{2}\int_{T_S}^{T_E} \|\sigma_P(t,T_E)\|^2 \, dt + \int_{T_S}^{T_E} \sigma_P(t,T_E)\, dW_t}, \\ P(T_S, T_E) &= \frac{P(t,T_E)}{P(t,T_S)} e^{-\frac{1}{2}\int_t^{T_S} \|\sigma_P(s,T_E)\|^2 - \|\sigma_P(s,T_S)\|^2 \, dt + \int_{T_S}^{T_E} \left(\sigma_P(s,T_E) - \sigma_P(s,T_S)\right)^\top dW_s}, \end{aligned} \tag{143}$$

where $\sigma_P(t,T) := \int_t^T \sigma_f(t,u)\, du$ and $\sigma_f(t,T)$ denote instantaneous volatility of $P(t,T)$ and $f(t,T)$, respectively.

Proof Since $\sigma_P(t,T)$ is a deterministic function in Guassian HJM, we have

$$\begin{aligned} \mathbb{E}\left[P(T_S,T_E)^{-1} \middle| \mathcal{F}_t \right] &= \frac{P(t,T_S)}{P(t,T_E)} \mathbb{E}\left[e^{\frac{1}{2}\int_t^{T_S} \|\sigma_P(s,T_E)\|^2 - \|\sigma_P(s,T_S)\|^2 \, dt - \int_{T_S}^{T_E} \left(\sigma_P(s,T_E) - \sigma_P(s,T_S)\right)^\top dW_s} \right] \\ &= \frac{P(t,T_S)}{P(t,T_E)} e^{\frac{1}{2}\int_t^{T_S} \|\sigma_P(s,T_E)\|^2 - \|\sigma_P(s,T_S)\|^2 \, dt + \frac{1}{2}\int_{T_S}^{T_E} \|\sigma_P(s,T_E) - \sigma_P(s,T_S)\|^2 \, ds} \\ &= \frac{P(t,T_S)}{P(t,T_E)} e^{\int_t^{T_S} \langle \sigma_P(s,T_E), \sigma_P(s,T_E) - \sigma_P(s,T_S) \rangle \, ds}. \end{aligned}$$

From bond reconstruction formula in (15), we see that in Gaussian FHJM model the volatility of $P(t,T)$ is $\sigma_P(t,T) = B_P(T-t)\mathbf{C}(t)$. Therefore,

$$\mathbb{E}\Big[P(T_S,T_E)^{-1}\Big|\mathcal{F}_t\Big] = \frac{P(t,T_S)}{P(t,T_E)} e^{\int_t^{T_S} B_P(T_E-s)^\top \left(B_P(T_E-s)-B_P(T_S-s)\right)\mathbf{C}(s)^\top \mathbf{C}(s)\,ds}. \quad (144)$$

Next, again using the fact that $\sigma_P(t,T)$ is a deterministic function, we have

$$\begin{aligned}\mathbb{E}\left[e^{\int_{T_S}^{T_E} r(u)\,du}\middle|\mathcal{F}_{T_S}\right] &= P(T_S,T_E)^{-1}\mathbb{E}\left[e^{\frac{1}{2}\int_{T_S}^{T_E}\|\sigma_P(t,T_E)\|^2\,dt+\int_{T_S}^{T_E}\sigma_P(t,T_E)\,dW_t}\right]\\ &= P(T_S,T_E)^{-1} e^{\int_{T_S}^{T_E}\|\sigma_P(t,T_E)\|^2\,dt} = P(T_S,T_E)^{-1} e^{\int_{T_S}^{T_E} B_P(T_E-s)^\top B_P(T_E-t)\mathbf{C}(t)^\top \mathbf{C}(t)\,dt}.\end{aligned} \quad (145)$$

The expression for $c^{\mathrm{ED}}(t)$ follows immediately from (144). Next, combining (145) and (144) and using tower property of conditional expectation, yields the expression for $c^{\mathrm{ON}}(t)$.

□

Proof of Theorem 3.4

Although the proof of Theorem 3.4 is very similar to that of Theorem 3.5, we provide it for completeness. Using Sepp and Rakhmonov (2023b) it is possible to show that function U, defined by

$$U(t) = U(t;X,Y,V) = c^{\mathrm{ON}}(\tau) e^{-\left(B_P(T_E-T_E\vee t)-B_P(T_S-T_S\vee t)\right)X-\left(\tilde{B}_P(T_E-T_E\vee t)-\tilde{B}_P(T_S-T_S\vee t)\right)Y},$$

where $c^{\mathrm{ON}}(t)$ defined, in turn, in (51), solves the problem as a functon of $\tau = T_E - t$:

$$-\partial_\tau U + \mathcal{L}U + \mathbb{1}_{\tau\le\Delta}\left(B(0)+\tilde{B}(0)\right)U = 0, \qquad U(0;X,Y,V) = 1, \quad (146)$$

where operator $\mathcal{L}$ is defined as follows:

$$\begin{aligned}\mathcal{L} &= \mathcal{L}^{(X)} + \mathcal{L}^{(Y)} + \mathcal{L}^{(V)}\\ \mathcal{L}^{(X)} &= \sum_{i=1}^d (\mathbf{D}X)^{(i)}\partial_{x_i} + \frac{V}{2}\sum_{i,j=1}^d \mathbf{M}_{ij}(T-\tau)\partial_{x_ix_j} + V\sum_{i=1}^d (\mathbf{C}(T-\tau)\beta)^{(i)}\partial_{vx_i},\\ \mathcal{L}^{(Y)} &= \sum_{i=1}^m (\tilde{\mathbf{D}}Y)^{(i)}\partial_{y_i} + V\sum_{i=1}^m \tilde{\Omega}^{(i)}_{T-\tau}\partial_{y_i},\\ \mathcal{L}^{(v)} &= [\kappa\theta - \kappa V]\partial_v + \frac{1}{2}\vartheta(T-\tau)^2 V\partial^2_{vv}.\end{aligned}$$

We assume the solution U is of the form

$$U(\tau) = \exp\left[B_1(\tau)^\top X + B_2(\tau)^\top Y + h_0(\tau) + h_1(\tau)v\right] \quad (147)$$

and substitute it into the PDE (146). To obtain the ODEs (61) for B_i and h_j, we collect the terms proportional to X, Y and V, respectively. Then initial conditions for ODEs (61) are set to reproduce conditions of the problem (146). Note that $c(t)$ does

not directly depend on X and Y. To demonstrate it, consider first $\tau \le \Delta$. After matching terms with X and Y, we see that $B_1(\tau)$, $B_2(\tau)$ solve the ODE

$$\frac{d}{d\tau}B_1 = B_1^\top \mathbf{D} + B(0), \quad B_1(0) = 0$$
$$\frac{d}{d\tau}B_2 = B_2^\top \tilde{\mathbf{D}} + \tilde{B}(0), \quad B_2(0) = 0,$$

which has solution

$$B_1(\tau) = B(0)\int_0^\tau e^{\mathbf{D}u}\,du, \quad B_2(\tau) = \tilde{B}(0)\int_0^\tau e^{\tilde{\mathbf{D}}u}\,du.$$

Hence the coefficients of X, Y in $c(t)$ equal, respectively

$$B(0)\int_{T_E-t}^{T_E} e^{\mathbf{D}u}\,du - B_P(T_E - t), \quad \tilde{B}(0)\int_{T_E-t}^{T_E} e^{\tilde{\mathbf{D}}u}\,du - \tilde{B}_P(T_E - t).$$

As the solutions of $B_P(\tau)$, $\tilde{B}_P(\tau)$ for the EP basis is given by $B_P(\tau) = B(0)\int_0^\tau e^{\mathbf{D}u}\,du$, $\tilde{B}_P(\tau) = \tilde{B}(0)\int_0^\tau e^{\tilde{\mathbf{D}}u}\,du$, those coefficients must vanish.

Next, when $\Delta < \tau \le T_E$ matching similarly the terms with X and Y, we see that $B_1(\tau)$, $B_2(\tau)$ solve the ODE

$$\frac{d}{d\tau}B_1 = B_1^\top \mathbf{D}, \quad B_1(0) = B_P(\Delta),$$
$$\frac{d}{d\tau}B_2 = B_2^\top \tilde{\mathbf{D}}, \quad B_2(0) = B_P(\Delta).$$

Now, the coefficients of X, Y in $c(t)$ equal, respectively

$$B_P(\Delta)e^{\mathbf{D}(T_S-t)} - \big(B_P(T_E - t) - B_P(T_S - t)\big), \quad \tilde{B}_P(\Delta)e^{\tilde{\mathbf{D}}(T_S-t)} - \big(\tilde{B}_P(T_E - t) - \tilde{B}_P(T_S - t)\big).$$

We observe that those coefficients also vanish, hence $c^{\mathrm{ON}}(t)$ does not depend on X, Y.

Proof of Theorem 3.5

Based on affine expansion framework for the moment-generating function, developed in Sepp and Rakhmonov (2023b) and applied in context of interest rate models in Sepp and Rakhmonov (2023a), we propose the following affine expansion for the convexity adjustment function in the log-normal SV dynamics (12).

Proposition 12.1 (Zeroth-order affine expansion with log-normal SV (12)) *The function $c(\tau, T; X, Y, v)$ in (51) can be decomposed as follows*

$$c^{\mathrm{ON}}(\tau; X, Y, v) = E^{[0]}(\tau; X, Y, v) + R^{[0]}(\tau; X, Y, v), \tag{148}$$

where $E^{[0]}(\tau; X, Y, v)$ is the leading term and $E^{[0]}(\tau; X, Y, v)$ is the remainder term. The leading term $E^{[0]}(\tau; X, Y, v)$ is given by the exponential-affine form:

$$E^{[0]}(\tau;v) = \exp\left[h_0(\tau) + h_1(\tau)v\right], \tag{149}$$

where h_0, h_1 satisfy the quadratic differential system (67) as a function of τ with the initial condition $B_1(0) = B_2(0) = 0$, $h_0(0) = h_1(0) = 0$.

The remainder term $R^{[0]}(\tau, T;X, Y, v)$ solves the following problem (we omit arguments for brevity):

$$\begin{aligned} -R^{[0]}_\tau + \mathcal{L}R^{[0]} &= -F^{[0]}(v, h_1, h_2)E^{[0]} \\ R^{[0]}(0, T;X, Y, v) &= 0, \end{aligned} \tag{150}$$

with operator $\mathcal{L}$ defined in Eq. (105) and residual function $F^{[0]}$

$$\begin{aligned} &F^{[0]}(v, h_1, h_2) = C^{(3)}(\tau;h_1, h_2)v^2, \\ &C^{(3)}(\tau) = \frac{1}{2}B_1^\top \mathbf{M}B_1 + h_1 B_1^\top \mathbf{C}\beta + B_2^\top \tilde{\Omega} - \kappa_2 h_1 + \frac{1}{2}\vartheta^2 h_1^2. \end{aligned} \tag{151}$$

Proof (Proof of Theorem 3.5) We consider FHJM–SV dynamics (9). Using Sepp and Rakhmonov (2023b) we can show that function U, defined by

$$U(\tau) = U(\tau;X, Y, v) = c^{\mathrm{ON}}(\tau)e^{-\left(B_P(T_E - T_E \vee t) - B_P(T_S - T_S \vee t)\right)X - \left(\tilde{B}_P(T_E - T_E \vee t) - \tilde{B}_P(T_S - T_S \vee t)\right)Y},$$

where $c^{\mathrm{ON}}(\cdot)$ is defined in (51), solves the problem as a functon of $\tau = T_S - t$

$$\begin{aligned} &-\partial_\tau U + \mathcal{L}U + \mathbb{1}_{\tau \le \Delta}\left(B(0) + \tilde{B}(0)\right)U = 0, \\ &U(0;X, Y, v) = 1, \end{aligned} \tag{152}$$

where operator $\mathcal{L}$ is defined as follows:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}^{(X)} + \mathcal{L}^{(Y)} + \mathcal{L}^{(v)} \\ \mathcal{L}^{(X)} &= \sum_{i=1}^{d} (\mathbf{D}X)^{(i)}\partial_{x_i} + \frac{(v+\theta)^2}{2}\sum_{i,j=1}^{d} \mathbf{M}_{ij}(T-\tau)\partial_{x_i x_j} + (v+\theta)^2 \sum_{i=1}^{d} (\mathbf{C}(T-\tau)\beta)^{(i)}\partial_{vx_i}, \\ \mathcal{L}^{(Y)} &= \sum_{i=1}^{m} (\tilde{\mathbf{D}}Y)^{(i)}\partial_{y_i} + (v+\theta)^2 \sum_{i=1}^{m} \tilde{\Omega}^{(i)}_{T-\tau}\partial_{y_i}, \\ \mathcal{L}^{(v)} &= \left[\hat{\kappa}_0(T-\tau) - \hat{\kappa}_1(T-\tau)v - \hat{\kappa}_2(T-\tau)v^2\right]\partial_v + \frac{1}{2}\vartheta(T-\tau)^2(v+\theta)^2\partial^2_{vv}, \end{aligned}$$

where we adopt the notation that $Z^{(i)}$ is the ith component of vector Z. Note that we introduced $\hat{\kappa}_0(t) = 0$, $\hat{\kappa}_1(t) = \kappa_1 + \kappa_2\theta$, $\hat{\kappa}_2(t) = \kappa_2$ to keep notations consistent.

We assume the solution U has the form

$$U(\tau) = \exp\left[B_1(\tau)^\top X + B_2(\tau)^\top Y + h_0(\tau) + h_1(\tau)v\right] \tag{153}$$

and substitute it into the PDE (152). To obtain the ODEs (67) for B_i and h_j, we match terms with X, Y and v^k, $k = 0, 1$, respectively. Then initial conditions for ODEs (67) are set to reproduce conditions of the problem (152). Using same arguments as in the proof of Theorem 3.4, we can show that $c^{\mathrm{ON}}(t)$ is independent of X and Y. □

Proof of Theorem 3.7

Proof For convenience, we consider shifted price process $\bar{F}_t := F_t + \Delta^{-1}$ that solves following SDE under risk-neutral measure $\mathbb{Q}$:

$$d\bar{F}_t = \bar{F}_t\sigma_t\Big(a_0(t)\,dW_t^{\mathbb{Q},(0)} + a_1(t)\,dw_t^{\mathbb{Q},(1)}\Big), \quad \bar{F}_0 = F + \Delta^{-1}$$
$$d\sigma_t = \big(\kappa_1 + \kappa_2\sigma_t\big)(\theta - \sigma_t)dt + \sigma_t\vec{\beta}(t)^\top dW_t^{\mathbb{Q},(0)} + \beta_0(t)\sigma_t dw_t^{\mathbb{Q},(1)}, \quad \sigma_0 = \sigma,$$

where $a_0(t)$, $a_1(t)$ are defined by (72). To derive conditions that ensure non-explosive dynamics for F_t, we use localization argument and introduce sequence of stopping times and corresponding stopped price processes

$$\tau_n = \inf\{t \geq 0 : \sigma_t > n\},$$
$$\bar{F}_t^{(n)} = \bar{F}_{t\wedge\tau_n} = \bar{F}_0 \exp\left[\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s\big(a_0(s)\,dW_s^{\mathbb{Q},(0)} + a_1(s)\,dw_s^{\mathbb{Q},(1)}\big)\right.$$
$$\left. -\frac{1}{2}\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s^2\big(\|a_0(s)\|^2 + a_1(s)^2\big)\,ds\right].$$

To show that $\bar{F}_t$ is a true martingale, the key step is to show that $\sup_n \mathbb{E}\Big[\bar{F}_t^{(n)}\log\bar{F}_t^{(n)}\Big] < \infty$. Denoting $M_t := \int_0^t 1_{s<\tau_n}\sigma_s\big(a_0(s)\,dW_s^{\mathbb{Q},(0)} + a_1(s)\,dw_s^{\mathbb{Q},(1)}\big)$, we have

$$\begin{aligned}\mathbb{E}\Big[\bar{F}_t^{(n)}\log\bar{F}_t^{(n)}\Big] &= \bar{F}_0\mathbb{E}\Big\{\exp\Big[M_t - \frac{1}{2}\langle M,M\rangle_t\Big]\Big[\log\bar{F}_0 + M_t - \frac{1}{2}\langle M,M\rangle_t\Big]\Big\}\\ &= \bar{F}_0\log\bar{F}_0 + \bar{F}_0\hat{\mathbb{E}}\Big\{M_t - \frac{1}{2}\langle M,M\rangle_t\Big\}.\end{aligned} \tag{154}$$

We note that in last equality we changed the measure from $\mathbb{Q}$ to $\hat{\mathbb{Q}}$, where

$$\frac{d\hat{\mathbb{Q}}}{d\mathbb{Q}}\bigg|_{\mathcal{F}_t} = \mathcal{E}\Big\{\exp\Big[\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s\big(a_0(s)\,dW_s^{\mathbb{Q},(0)} + a_1(s)\,dw_s^{\mathbb{Q},(1)}\big) - \frac{1}{2}\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s^2\big(\|a_0(s)\|^2 + a_1(s)^2\big)\,ds\Big]\Big\}.$$

Hence, by Girsanov theorem, the pair

$$\hat{W}_t^{(0)} := W_t^{\mathbb{Q},(0)} - \int_0^t \sigma_u a_0(u)\mathbb{1}_{u<\tau_n}\,du, \qquad \hat{w}_t^{(1)} := w_t^{\mathbb{Q},(1)} - \int_0^t \sigma_u a_1(u)\mathbb{1}_{u<\tau_n}\,du$$

defines two-dimensional Brownian motions under $\hat{\mathbb{Q}}$. Thus, the right-hand side of (154) equals

$$\begin{aligned}\bar{F}_0\log\bar{F}_0 + \bar{F}_0\hat{\mathbb{E}}\Big\{&\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s\big(a_0(s)\,d\hat{W}_s^{(0)} + a_1(s)\,d\hat{W}_s^{(1)}\big)\\ &+\frac{1}{2}\int_0^t \mathbb{1}_{s<\tau_n}\sigma_s^2\big(\|a_0(s)\|^2 + a_1(s)^2\big)\,ds\Big\}.\end{aligned} \tag{155}$$

It follows that the right-hand side of Eq. (155) is uniformly bounded, if $\mathbb{E}\sigma_t^2 < \infty$ for $t \le T^*$ and process σ_t does not blow up in $\hat{\mathbb{Q}}$. The first condition is satisfied if $\kappa_2 > 0$ and κ_1, θ satisfy conditions in (13), see Sepp and Rakhmonov (2023b). To check the second condition, we see that under $\hat{\mathbb{Q}}$ the volatility process σ_t solves, up to time τ_n, the SDE

$$\begin{aligned} d\sigma_t &= \big(\kappa_1 + \kappa_2\sigma_t\big)(\theta - \sigma_t)dt + \sigma_t\vec{\beta}(t)^\top d\hat{W}_t^{(0)} + \beta_0(t)\sigma_t d\hat{W}_t^{(1)} + \sigma_t^2\big[\beta(t)^\top a_0(t) + \beta_0(t)a_1(t)\big]\,dt \\ &= \Big[\kappa_1\theta - (\kappa_1 - \kappa_2\theta)\sigma_t - \Big(\kappa_2 - \vec{\beta}(t)^\top a_0(t) - \beta_0(t)a_1(t)\Big)\sigma_t^2\Big]\,dt + \sigma_t\vec{\beta}(t)^\top d\hat{W}_t^{(0)} + \beta_0(t)\sigma_t d\hat{W}_t^{(1)}. \end{aligned} \tag{156}$$

Its solution σ_t does not explode in $\hat{\mathbb{Q}}$ if quadratic term in the drift remains positive. Combining both, we see that if (76) is satisfied, then $\sup_n \mathbb{E}\left[\bar{F}_t^{(n)} \log \bar{F}_t^{(n)}\right] < \infty$ as $\tau_n \uparrow +\infty$, $n \to +\infty$. Using de la Vallée–Poussin criterion for uniform integrability, we conclude that family $\left\{\bar{F}_t^{(n)}\right\}_{n=1}^{\infty}$ is uniformly integrable, hence

$$\mathbb{E}[\bar{F}_t|\mathcal{F}_s] = \bar{F}_s, \quad 0 \le s \le t \le T^*. \tag{157}$$

□

Appendix 5: Second-order affine expansion

Theorem 13.1 (*Second-order affine expansion*) *The MGF function defined in Eq.* (106) *can be decomposed as follows*:

$$G(\tau, T, z, v; \phi) = \exp\{-\phi z\}\left[E^{[2]}(\tau, T, v; \phi) + R^{[2]}(\tau, T, v; \phi)\right], \tag{158}$$

where $E^{[2]}$ *is the leading term and* $R^{[2]}$ *is the remainder term.*

The leading term $E^{[2]}(\tau, T; z, v)$ *is given by the exponential-affine form*:

$$E^{[2]}(\tau, T, v; \phi) = \exp\left\{\sum_{k=0}^{4} A^{(k)}(\tau, T; \phi)v^k\right\}, \tag{159}$$

where vector function $\mathbf{A}(\tau) = \left\{A^{(k)}(\tau, T; \phi)\right\}$, $k = 0, 1, \ldots, 4$, *solves the following quadratic differential system as a function of* τ:

$$\begin{aligned} A_\tau^{(k)} &= \mathbf{A}^\top M^{(k)}\mathbf{A} + \left(L^{(k)}\right)^\top \mathbf{A} + H^{(k)}, \\ M^{(k)} &= \left\{\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\theta^2\vartheta^2}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \theta\vartheta^2 & \theta^2\vartheta^2 & 0 & 0 \\ 0 & \theta^2\vartheta^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\vartheta^2}{2} & 2\theta\vartheta^2 & \frac{3}{2}\theta^2\vartheta^2 & 0 \\ 0 & 2\theta\vartheta^2 & 2\theta^2\vartheta^2 & 0 & 0 \\ 0 & \frac{3}{2}\theta^2\vartheta^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \right. \\ &\quad \left. \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\vartheta^2}{2} & 2\theta\vartheta^2 & \frac{3}{2}\theta^2\vartheta^2 & 0 \\ 0 & 2\theta\vartheta^2 & 2\theta^2\vartheta^2 & 0 & 0 \\ 0 & \frac{3}{2}\theta^2\vartheta^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \right\}, \end{aligned} \tag{160}$$

$$\left.\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \vartheta^2 & 3\theta\vartheta^2 & 2\theta^2\vartheta^2 \\ 0 & \vartheta^2 & 4\theta\vartheta^2 & 3\theta^2\vartheta^2 & 0 \\ 0 & 3\theta\vartheta^2 & 3\theta^2\vartheta^2 & 0 & 0 \\ 0 & 2\theta^2\vartheta^2 & 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{2}\vartheta^2 & 4\theta\vartheta^2 \\ 0 & 0 & 2\vartheta^2 & 6\theta\vartheta^2 & 4\theta^2\vartheta^2 \\ 0 & \frac{3}{2}\vartheta^2 & 6\theta\vartheta^2 & \frac{9}{2}\theta^2\vartheta^2 & 0 \\ 0 & 4\theta\vartheta^2 & 4\theta^2\vartheta^2 & 0 & 0 \end{pmatrix}\right\},$$

$$L^{(k)} = \left\{\begin{pmatrix} 0 \\ \hat{\kappa}_0 - \theta^2(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi \\ \theta^2\vartheta^2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -\hat{\kappa}_1 - 2\theta(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi \\ 2(\hat{\kappa}_0 + \theta\vartheta^2 - \theta^2(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \\ 3\theta^2\vartheta^2 \\ 0 \end{pmatrix}, \right.$$

$$\begin{pmatrix} 0 \\ -(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi - \hat{\kappa}_2 \\ \vartheta^2 - 2\hat{\kappa}_1 - 4\theta(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi \\ 3(\hat{\kappa}_0 + 2\theta\vartheta^2 - \theta^2(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \\ 6\theta^2\vartheta^2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ -2\hat{\kappa}_2 - 2(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi \\ 3(-\hat{\kappa}_1 + \vartheta^2 - 2\theta(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \\ 4(\hat{\kappa}_0 + 3\theta\vartheta^2 - \theta^2(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \end{pmatrix},$$

$$\left.\begin{pmatrix} 0 \\ 0 \\ 0 \\ -3(\hat{\kappa}_2 + (\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \\ 2(-2\hat{\kappa}_1 + 3\vartheta^2 - 4\theta(\langle\hat{a}_0,\beta\rangle + \beta_0\hat{a}_1)\phi) \end{pmatrix}\right\},$$

$$H^{(k)} = \left\{\frac{1}{2}\theta^2(\|\hat{a}_0\|^2 + \hat{a}_1^2)\phi^2 + \theta^2\phi\hat{b},\ \theta(\|\hat{a}_0\|^2 + \hat{a}_1^2)\phi^2 + 2\theta\phi\hat{b},\ \frac{1}{2}(\|\hat{a}_0\|^2 + \hat{a}_1^2)\phi^2 + 2\phi\hat{b}, 0, 0\right\},$$

with the initial condition $\mathbf{A}(0) = (0,0,0,0,0)^\top$, *where we omit argument* $T-\tau$ *in functions* $\hat{\kappa}_i, \hat{a}_i, \hat{b}, \vec{\beta}, \beta_0, \vartheta$. *The remainder term* $R^{[2]}(\tau, T, v; \phi)$ *solves the following problem*:

$$\begin{aligned} -R^{[2]}_\tau + \mathcal{L}^{(v)}R^{[2]} &= -F^{[2]}(v, A^{(1)}, A^{(2)})E^{[2]}(\tau, T, v; \phi) \\ R^{[2]}(0, T, v; \phi) &= 0, \end{aligned} \tag{161}$$

with operator $\mathcal{L}^{(v)}$ *defined in Eq.* (105) *and residual function* $F^{[2]}$ *given by*

$$
\begin{aligned}
F^{[2]}(v, A^{(1)}, A^{(2)}, A^{(3)}, A^{(4)}) &= \sum_{n=5}^{8} C^{(n)}(\tau; A^{(1)}, A^{(2)}, A^{(3)}, A^{(4)}) v^n, \\
C^{(5)}(\tau) &= \left(6A^{(2)}A^{(3)} + 9q(A^{(3)})^2 + 4A^{(1)}A^{(4)} + 4q(4A^{(2)} + 3qA^{(3)})A^{(4)}\right)\vartheta^2 - \\
&\quad - 4A^{(4)}(\langle \hat{a}_0, \beta \rangle + \beta_0 \hat{a}_1)\phi - 4A^{(4)}\kappa_2, \\
C^{(6)}(\tau) &= \left(\frac{9}{2}(A^{(3)})^2 + 24qA^{(3)}A^{(4)} + 8A^{(4)}(A^{(2)} + q^2 A^{(4)})\right)\vartheta^2, \\
C^{(7)}(\tau) &= \left(4A^{(4)}(3A^{(3)} + 4qA^{(4)})\right)\vartheta^2, \\
C^{(8)}(\tau) &= 8(A^{(4)})^2\vartheta^2.
\end{aligned}
\tag{162}
$$

References

Andersen, L. B., & Lake, M. (2019). Robust high-precision option pricing by Fourier transforms: Contour deformations and double-exponential quadrature. *Wilmott Magazine, 102*, 22–41.

Andersen, T. G., & Lund, J. (1997). Estimating continuous-time stochastic volatility models of the short-term interest rate. *Journal of Econometrics, 77*(2), 343–377.

Andersen, L. B., & Piterbarg, V. V. (2007). Moment explosions in stochastic volatility models. *Finance and Stochastics, 11*(1), 29–50.

Andersen, L. B., & Piterbarg, V. V. (2010). *Interest rate modeling*. Atlantic Financial Press.

Bernard, C., Cui, Z., & McLeish, D. (2017). On the martingale property in stochastic volatility models based on time-homogeneous diffusions. *Mathematical Finance, 27*(1), 194–223.

Brigo, M., & Mercurio, F. (2007). *Interest–rate models: Theory and practice*. Springer.

Carr, P., Fisher, T., & Ruf, J. (2014). On the hedging of options on exploding exchange rates. *Finance and Stochastics, 18*, 115–144.

Cheyette, O. (1992) Markov representation of the Heath–Jarrow–Morton Model. Working paper, BARRA.

Christensen, J. H., Diebold, F. X., & Rudebusch, G. D. (2009). An arbitrage-free generalized Nelson–Siegel term structure model. *Econometrics Journal, 12*, C33–C64.

Christensen, J. H., Diebold, F. X., & Rudebusch, G. D. (2011). The affine arbitrage-free class of Nelson-Siegel term structure models. *Journal of Econometrics, 164*(1), 4–20.

Collin-Dufresne, P., & Goldstein, R. S. (2002). Do bonds span the fixed income markets? Theory and evidence for unspanned stochastic volatility. *The Journal of Finance, 57*(4), 1685–1730.

Diebold, F. X., & Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of Econometrics, 130*(2), 337–364.

Duchitskii, I. & Piterbarg, V. (2024). Estimating mean reversions in interest rate models. Available at SSRN: https://ssrn.com/abstract=4897350

Filipović, D. (1999). A note on the Nelson–Siegel Family. *Mathematical Finance, 9*(4), 349–359.

Filipović, D. (2000). Exponential-polynomial families and the term structure of interest rates. *Bernoulli, 1*, 1081–1107.

Filipović, D. (2023). Discount models. *Finance and Stochastics, 27*(4), 933–946.

Glasserman, P., & Kim, K.-K. (2010). Moment explosions and stationary distributions in affine diffusion models. *Mathematical Finance, 20*(1), 1–33.

Heath, D., Jarrow, R., & Morton, A. (1992). Bond pricing and the term structure of interest rates: A new methodology for contingent claims valuation. *Econometrica: Journal of the Econometric Society, 1*, 77–105.

Henrard, M. (2005). Eurodollar futures and options: Convexity adjustment in HJM one-factor model. Available at SSRN 682343.

Hunt, P., & Kennedy, J. (2004). *Financial derivatives in theory and practice* (Vol. 556). Wiley.

Jäckel, P., & Kawai, A. (2005). The future is convex. Wilmott February, 2–13.

Jacquier, A., & Oumgari, M. (2023). Interest rate convexity in a Gaussian framework. Working paper.

Karasinski, P., & Sepp, A. (2012). Beta stochastic volatility model. *Risk Magazine, 1*, 66–71.
Kirikos, G., & Novak, D. (1997). Convexity conundrums. *Risk, 1*, 60–61.
Lewis, A. L. (2000). *Option valuation under stochastic volatility*. Finance Press.
Lions, P. L., & Musiela, M. (2007). Correlations and bounds for stochastic volatility models. *Annales de l'Institut Henri Poincaré C, Analyse non linéaire, 24*(1), 1–16.
Lipton, A. (2001). *Mathematical methods for foreign exchange: A financial engineer's approach*. World Scientific.
Lucic, V. (2004). Forward-start options in stochastic volatility models. *The best of Wilmott, 1*, 413–420.
Lucic, V. (2016) Non-parametric local volatility swaption formula revisited. Available at SSRN: https://ssrn.com/abstract=2808470
Lyashenko, A., & Goncharov, Y. (2023). Introducing the factor HJM term structure modeling framework. *Risk Magazine, 2023*, 1–6.
Lyashenko, A. & Mercurio, F. (2019) Looking forward to backward-looking rates: A modeling framework for term rates replacing LIBOR. Available at SSRN: https://ssrn.com/abstract=2987832
Mercurio, F. (2017) The present of futures: valuing eurodollar-futures convexity adjustments in a multi-curve world. Available at SSRN: https://ssrn.com/abstract=2987832
Michalski, K. A., & Mosig, J. R. (2016). Efficient computation of Sommerfeld integral tails-methods and algorithms. *Journal of Electromagnetic Waves and Applications, 30*(3), 281–317.
Musiela, M. (1993). *Stochastic PDE's and term structure models*. University of New South Wales.
Nelson, C., & Siegel, A. (1987). Parsimonious modeling of yield curves. *Journal of Business, 60*, 473–489.
Piterbarg, V., & Renedo, M. (2006). Eurodollar futures convexity adjustments in stochastic volatility models. *Journal of Computational Finance, 9*, 71–94.
Ruf, J. (2015). The martingale property in the context of stochastic differential equations. *Electronic Communications in Probability, 20*, 1–10.
Sepp, A., & Rakhmonov, P. (2023a). A robust stochastic volatility model for interest rates. *Risk Magazine, 2023*, 1–6.
Sepp, A., & Rakhmonov, P. (2023b). Log-normal stochastic volatility model with quadratic drift. *International Journal of Theoretical and Applied Finance, 26*(08), 2450003.
Sepp, A. & Rakhmonov, P. (2025). Risk-premia and measure change in factor HJM model with stochastic volatility. Available at SSRN: https://ssrn.com/abstract=5161294
Sin, C. A. (1998). Complications with stochastic volatility models. *Advances in Applied Probability, 30*(1), 256–268.
Skov, J. B., & Skovmand, D. (2021). Dynamic term structure models for SOFR futures. *Journal of Futures Markets, 41*(10), 1520–1544.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.